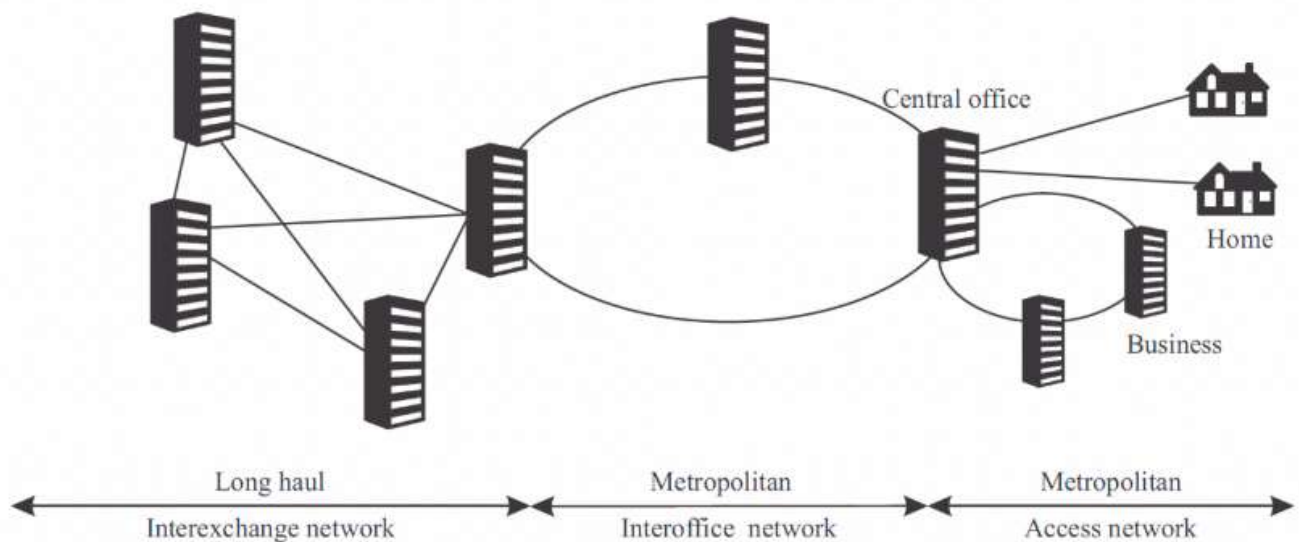


Theory of Transport Networks - 24_25

Author: Dr. Plumcake

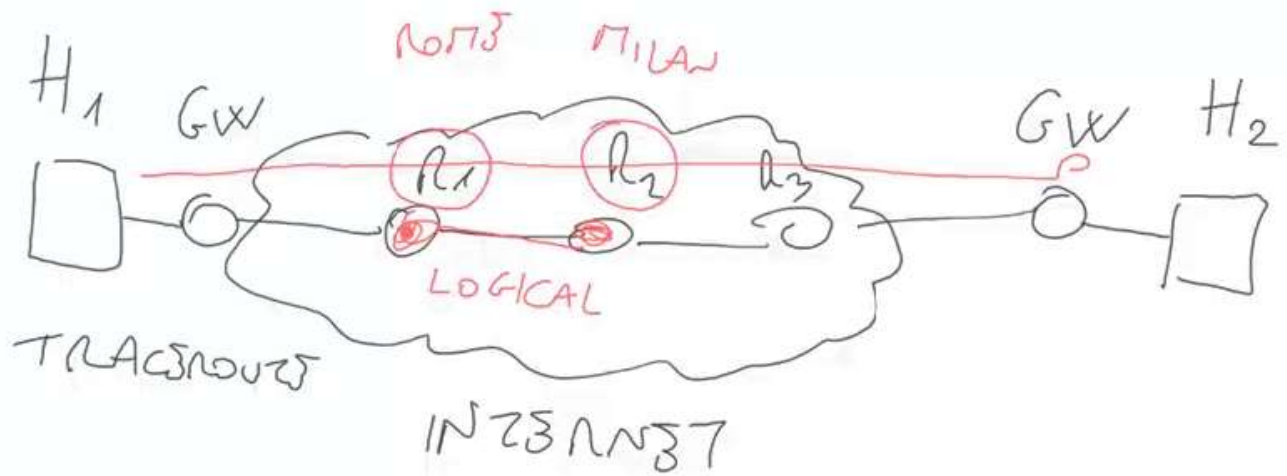
Master's Degree in Cybersecurity - Network Infrastructures

Introduction to Optical Networks



The goal of a Transport Network (TN) is to provide connectivity to clients requesting a connection. The TN can be divided into three parts:

1. **Metropolitan Network:** This part lies within a large city or region and includes:
 1. **Access Network:** The connection typically begins in the Access Network Area, where users send and receive signals via their cables (this portion of the network is covered in another part of the course) to a central office (*sometimes POP = Point of Presence*). The connection then traverses two additional "layers" of the network's general infrastructure:
 2. **Inter-office Network:** Usually organized with a Ring Topology (to mitigate the impact of network malfunctions), this layer aggregates and directs user accesses to the deeper parts of the network.
2. **Inter-Exchange Network (Long Haul):** At this level, the Transport Network itself is employed, utilizing various technologies to transport signals to their destinations. The majority of the infrastructure relies on Optical Cables, as the bandwidth required to handle the vast amounts of data is extremely high.



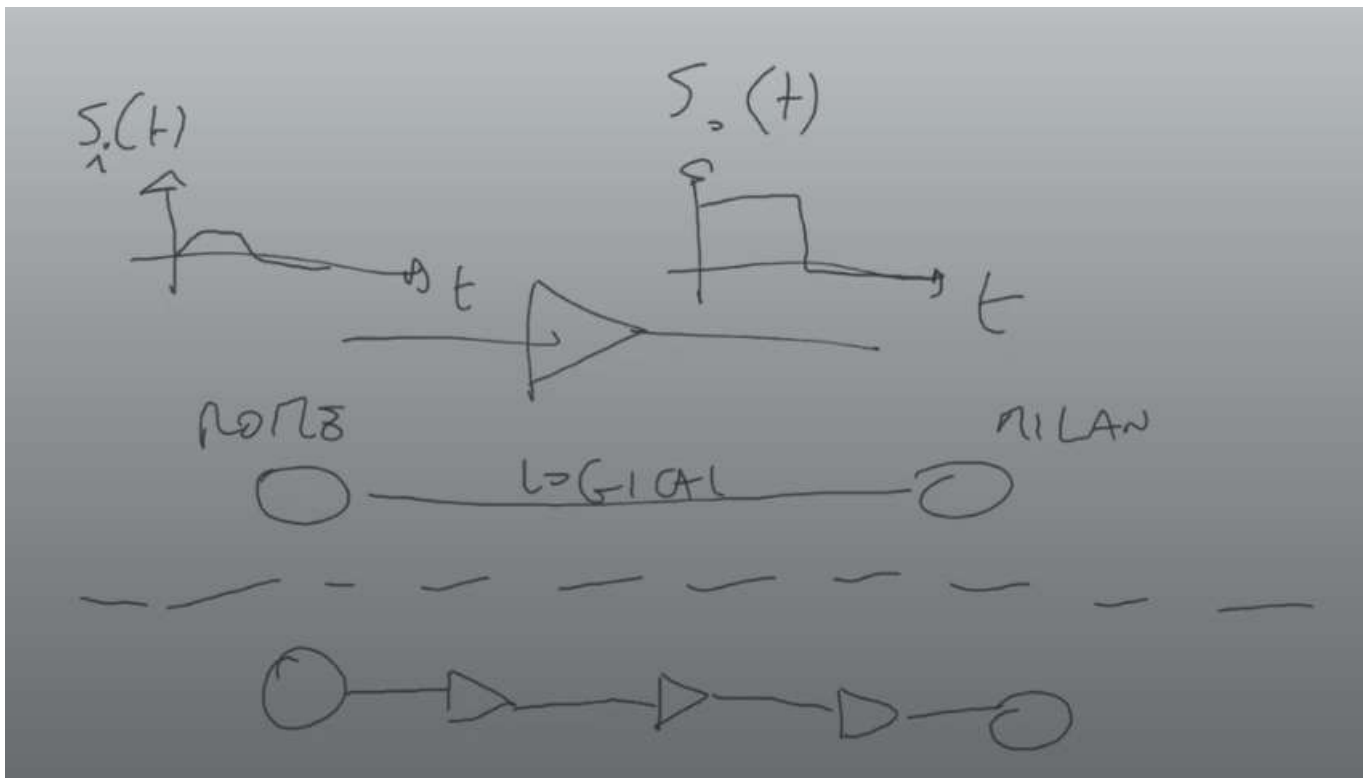
Logical vs. Physical Connections

The challenge of connecting two hosts, or two routers, in a transport network lies in establishing a physical connection. A key distinction must be made:

1. **A Logical Connection** Enables two devices to communicate through the network without detailing how the connection is structured. This type of connection is often represented as a cloud/network, symbolizing that the *actual path* from point A to point B is irrelevant. Logical connections are more commonly used when dealing with abstract network layers, such as Layer 3 (IP, Network, Connectivity, etc.).
2. **A Physical Connection** Represents a concrete path between devices, following a series of optical cables to the destination. The path from the signal's source/client to its destination is referred to as a *light-path*. This representation is typically used when dealing with lower layers of the network (Layer 1 or 2).

Signal Quality and Amplification

Another issue in connecting two devices is signal degradation along the path. The solution to this problem is an *Amplifier*: a device that takes a degraded signal as input, restores its original quality by amplifying it, and then sends it as output.



Connection Creation

Connections are not established by the network's clients but by the *Carrier/Providers*. When a client wishes to connect to another endpoint in the network, it sends a connection creation request to the provider. The provider then constructs the communication path between the two clients through the transport network.

Types of Network Services

Network services can be categorized into two types:

- **Connection-less:** The source sends messages whenever data is available (e.g., email).
- **Connection-oriented:** An agreement must exist between the sender and receiver before the connection is established. This type of service often requires an agreement with the provider as well, since the provider determines the maximum bandwidth available for the connection. This arrangement is useful to prevent channel overflow, as the connection's bandwidth limit is predefined through the provider's agreement.

Switching Paradigms in Core Devices

Core network devices operate according to one of the following switching paradigms:

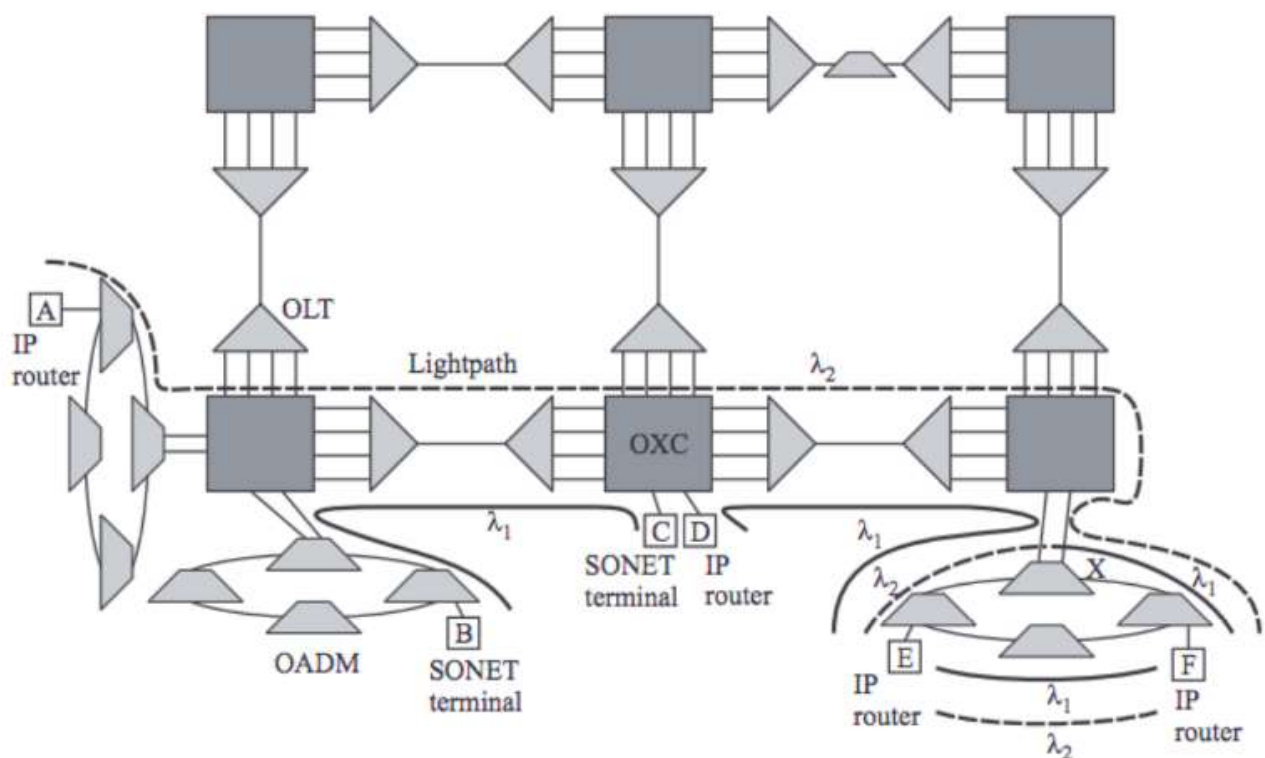
- **Circuit Switching:** Relies on static multiplexing, dividing a channel into sub-channels and redirecting connections through these sub-channels (using switching tables when

necessary). This approach is deterministic but may be inefficient in resource utilization, as unused sub-channels can lower the network's overall performance.

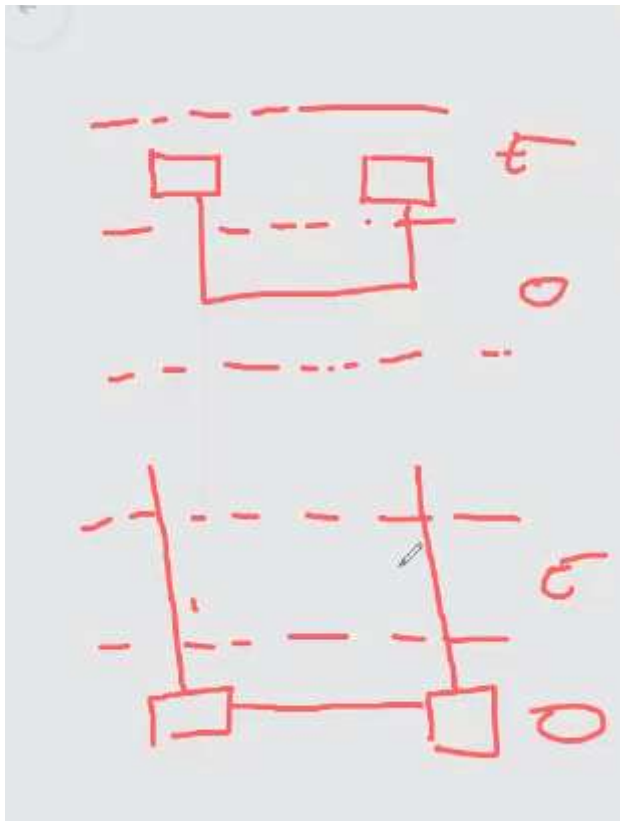
- **Packet Switching:** Relies on statistical multiplexing, following a "first-come, first-served" principle that allows packets to utilize the channel's full capacity. However, this approach is not deterministic. If packets arrive simultaneously, delays can occur, and there is no guaranteed upper limit on the waiting time for packets in the queue (in some cases, indefinite delays may occur).

Second Generation Optical Networks

Optical Networks can be divided into two generations:



1. **First Generation:** In this generation, optics are primarily used for transmission and to provide capacity. Switching and other complex functions are handled by electronics. When the network needs to perform a switching operation or similar tasks, the signals are transferred from the optical layer to the electronic domain for processing and then returned to the optical layer.
2. **Second Generation:** In this generation, routing, switching, and intelligence operations are executed entirely within the optical layer. Signals are transferred to and from the electronic domain only at the beginning and end of transmission.



Multiplexing Techniques

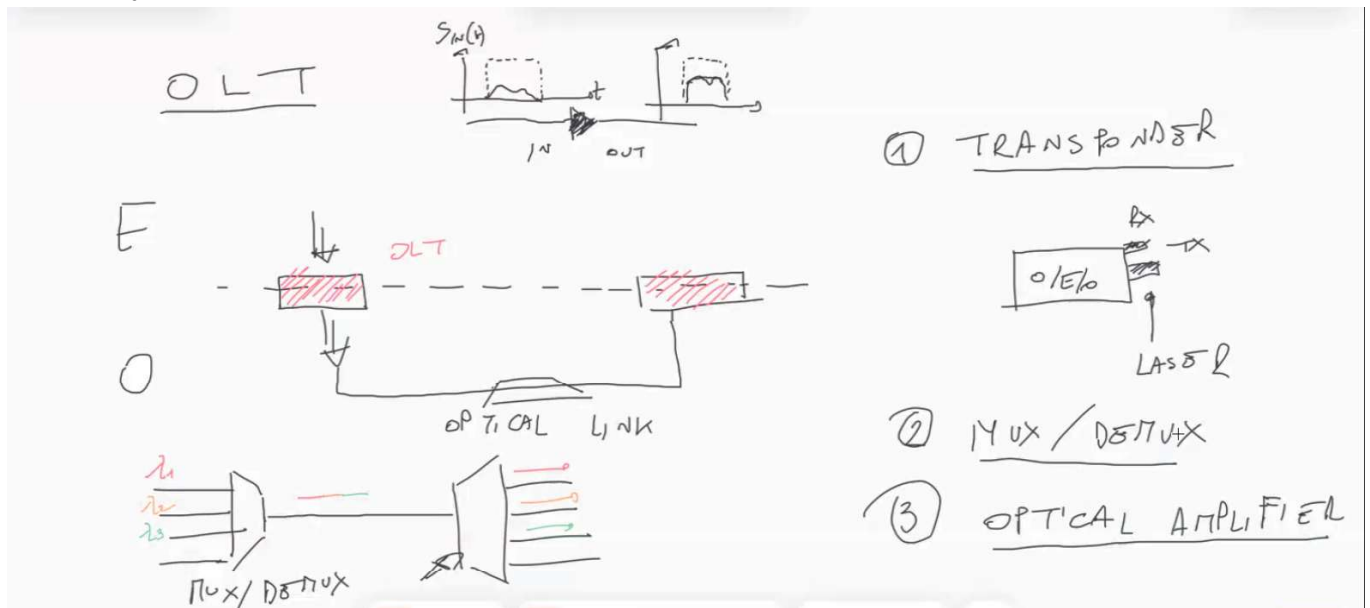
Multiplexing increases the transmission capacity of a fiber by aggregating multiple signals. These techniques are also employed in Passive Optical Networks (PON).

- **TDM (Time Division Multiplexing):** Multiple data streams from individual users or sources (*tributary flows*) are combined (*multiplexed*) into a single data stream. This is achieved by transmitting each tributary flow in a distinct "time slot" within a time window of specific duration. The total transmission capacity must equal $N \times R$, where N is the number of tributary flows and R is the average capacity of each flow. However, this technique faces practical limitations when dealing with too many users or large tributary flows.
- **WDM (Wavelength Division Multiplexing):** This technique exploits the wavelength property of optical signals. Each optical signal has a wavelength ($\lambda = c \div f$, where c is the speed of light and f is the signal's frequency) related to its frequency and, visually, its color. For example, different colors of light correspond to different wavelengths. WDM transmits signals simultaneously on different wavelengths. This requires devices at the ends of optical cables to distinguish signals based on their wavelength. Multiplexed signals are typically split into different wavelengths using physical prism-like devices. Compared to TDM, WDM is resource-efficient because it doesn't require increasing channel capacity while scaling the data rate.

Components of a Second-Generation Optical Network

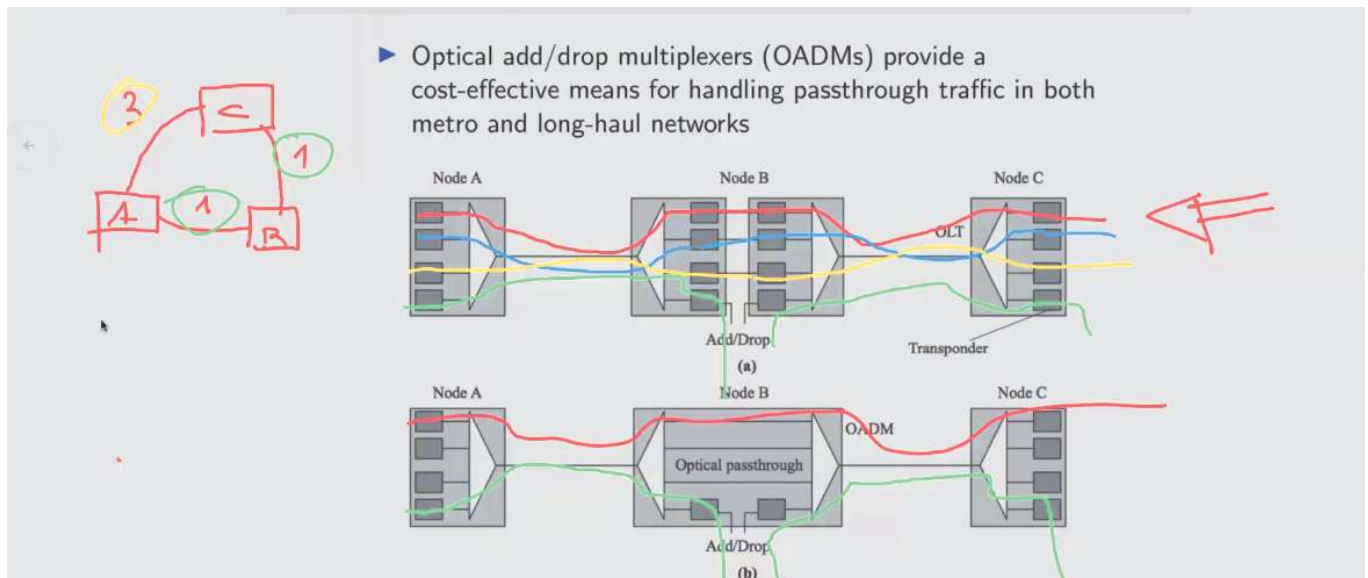
The main multiplexing technique in second-generation optical networks is WDM. The primary device for switching signals is the **Optical Cross-Connect (OXC)**, which has multiple ports and reconfigurable functionality to redirect optical signals to different paths.

Other key devices include:

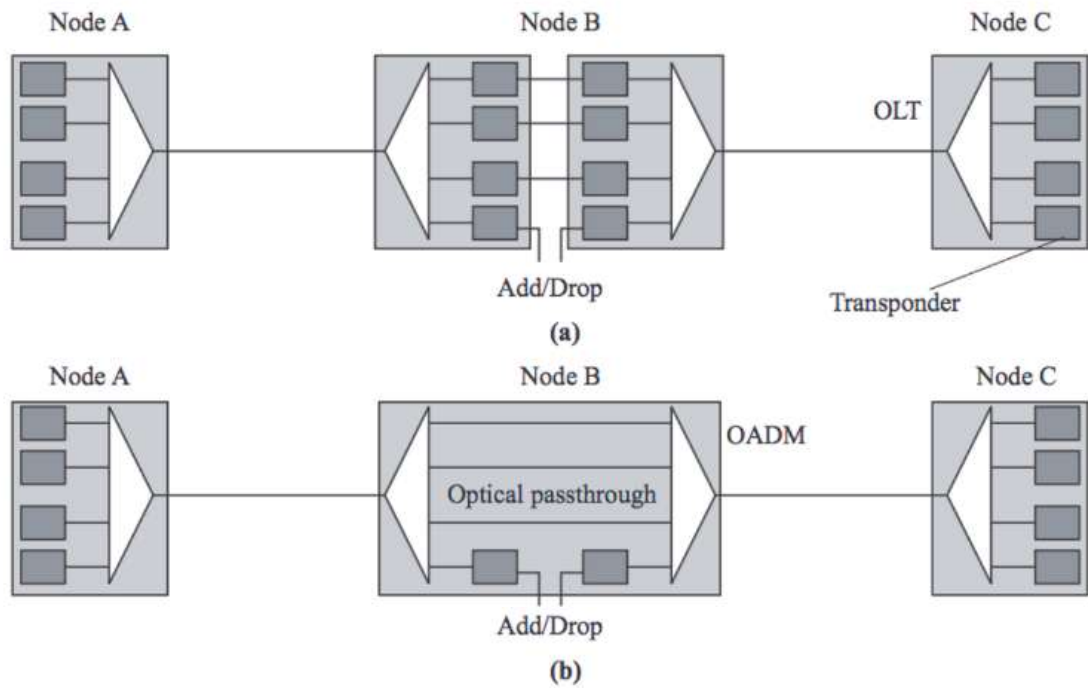


- **OLT (Optical Line Terminal):** Represented as a triangle in diagrams, its main job is to multiplex or de-multiplex incoming wavelengths. It has four ports on one side and one port on the other.
 - When a multiplexed signal arrives, the OLT splits it into individual signals, distributing them across the four ports. Conversely, multiple signals arriving on the four ports are multiplexed into one signal for output.
 - The OLT comprises three functional components:
 1. **Transponder:** Converts incoming light signals from clients into electrical signals and vice versa. Key functions include:
 - Adjusting wavelengths incompatible with the optical network (e.g., converting wavelengths to fit the $1.55 \mu m$ window).
 - Adding OTN Overhead, which encapsulates signals with additional information to enable network services (similar to IP routing).
 - Checking the signal's Bit Error Rate (BER) when in the electrical domain, enabling more efficient error detection through encoding techniques.
 - Structural decisions are often made to avoid overusing transponders due to their high cost.
 2. **Wavelength Multiplexer:** Multiplexes or de-multiplexes signals by separating or combining wavelengths on different cables.
 3. **Optical Amplifier:** Boosts signal power to rejuvenate degraded signals.

OLTs are typically deployed at both ends of a point-to-point optical link to convert electrical signals into optical signals and vice versa.

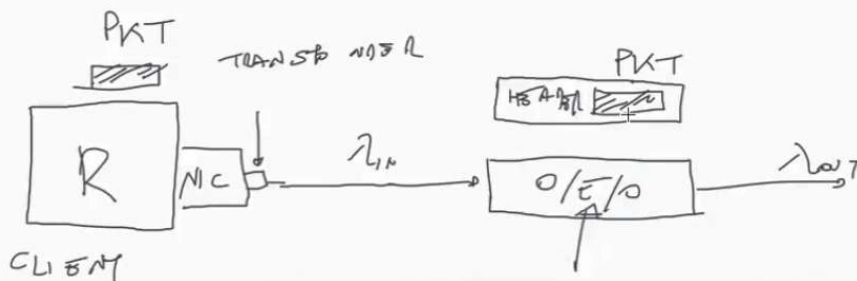


- **OADM (Optical Add/Drop Multiplexer):** Represented as a trapezium in diagrams, this device adds or removes signals from a multiplexed transmission.
 - It has four ports: one for the input multiplexed signal, one for the output multiplexed signal, and two additional ports for adding and dropping signals.
 - The OADM de-multiplexes the incoming signal, removes the designated wavelength, and adds a new signal (even with the same wavelength) before multiplexing and transmitting the output.
 - A distinguishing feature is that OADMs do not require additional transponders for signals not being added or dropped; optical passthrough cables handle those signals. This reduces costs.
 - However, OADMs are not reconfigurable, meaning the wavelengths for adding or dropping are fixed.



(1) $\lambda_{in} \in 1,3 \mu m \Rightarrow \lambda_{out} \in 1,55 \mu m$

(2) ADD OVERHEAD



Managing Wavelength Constraints and Blocking Events

A constraint of remaining in the optical domain is the limited number of wavelengths available for optical signals. It is important that two clients use different wavelengths; otherwise, their signals will be indistinguishable from one another. This limitation can be bypassed for a limited portion of the path using *Wavelength Conversion*: an operation that changes the color (wavelength) of an optical signal. Visually, this can be thought of as altering the light's color.

Wavelength conversion is only possible in the electrical domain, so if the signal is in the optical domain, it must first be moved to the electrical domain and then back to the optical domain. A

transponder can perform wavelength conversion, as it transitions signals between the optical and electrical domains and vice versa. This process is crucial for addressing the possibility of a *Blocking Event*: a situation in which a customer sends a connection request, but the request is rejected because no available wavelength exists for that customer.

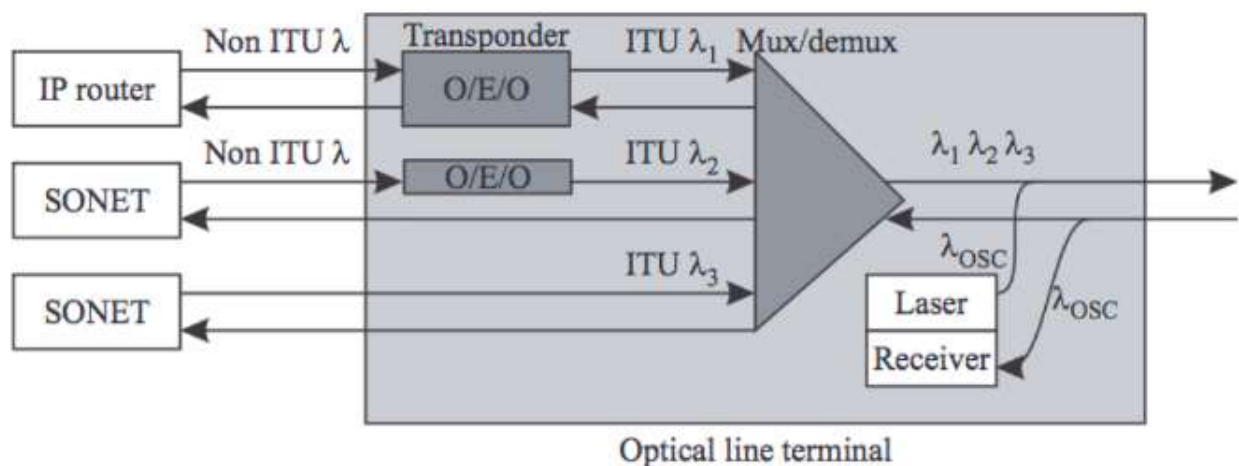
To avoid a blocking event, there are two possible solutions:

1. **Find Another Path:** Routing becomes essential in transport networks. If the most direct path does not have available wavelengths, an alternative path may exist that provides a wavelength not in use for the destination.
2. **Wavelength Conversion:** This operation temporarily transforms the signal's wavelength, enabling it to traverse the most direct path, which would otherwise be incompatible. However, this solution has drawbacks, including the monetary expense of acquiring devices capable of performing wavelength conversion.

In terms of network structure, core networks typically have a high concentration of OXCs to efficiently redirect optical signals, supported by a mesh topology. As we move toward the edge of the network, the topology transitions to a ring structure for enhanced stability, and OADMs become more prevalent.

An additional property of the network is *Transparency*: Clients cannot perceive signals not addressed to them.

Resuming OLT Topic



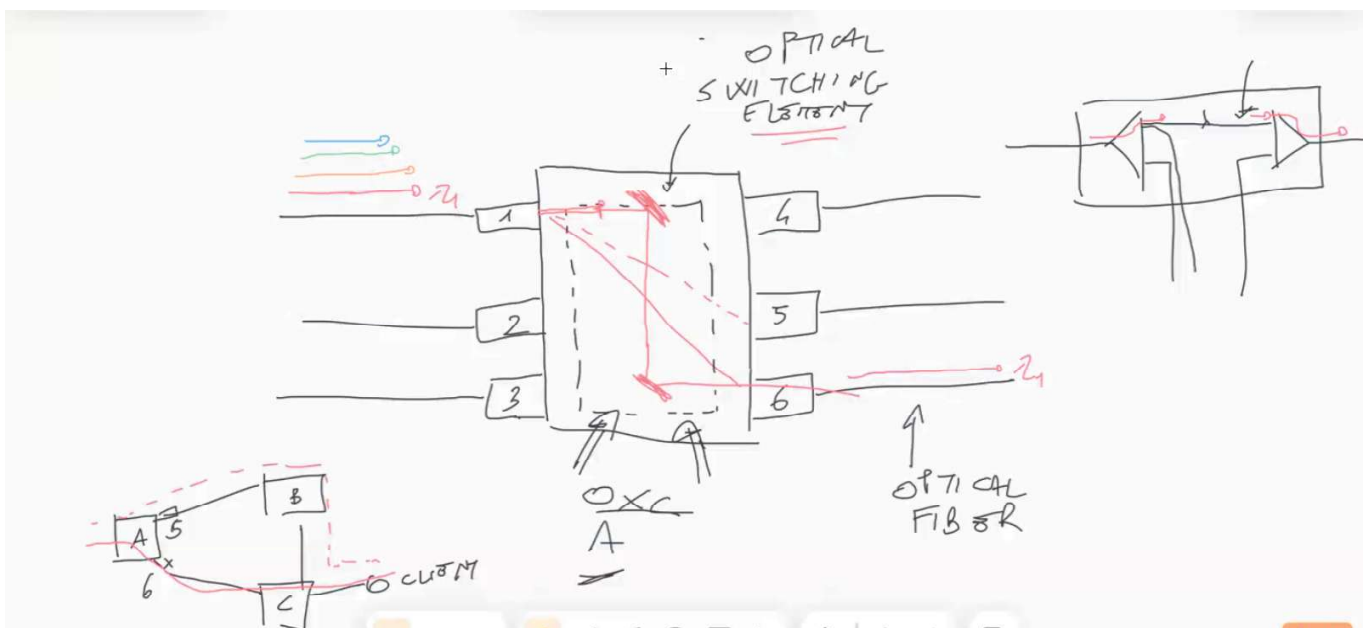
The OLT does not always include all three of its main components. Depending on the situation, it may be equipped with fewer components.

The optical network is a heterogeneous environment, where devices are often expensive, and older equipment may still be operational within the same network. In the first generation of optical networks, wavelengths fell into a different window than those used today due to technological limitations. Specifically, non-ITU light signals originated from LED light sources, while ITU-compliant signals are generated by lasers. The transponder's role in adapting incoming signals to the ITU standard includes:

- **Centering the Signal:** The signal is adjusted from an input wavelength, λ_{in} $1.3 \mu m$, to an output wavelength, λ_{out} $1.5 \mu m$, potentially altering or maintaining the signal's original color.
- **Adding Overhead:** This operation occurs in the electrical domain. The transponder recovers the packet or original message from the optical domain, adds a header, and reconverts the signal to the optical domain for further transmission.

The network is monitored by a dedicated transponder equipped with a laser that sends optical signals at a specific wavelength, reserved solely for monitoring purposes. This channel is called the *Optical Supervisory Channel* and is typically associated with the OLT.

OXC: Optical Switching Element

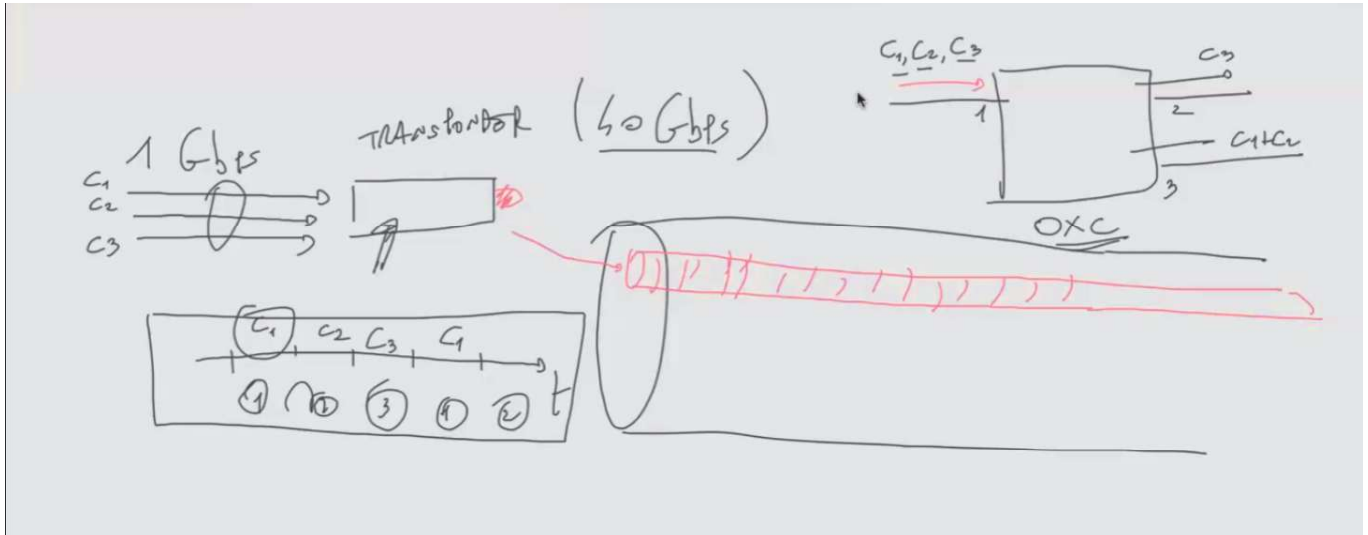


The OXC (Optical Cross-Connect) is a device with multiple ports, each capable of transmitting optical signals. Its primary function is to redirect and switch these signals along different paths. An OXC de-multiplexes an incoming transmission at one port, isolates a specific wavelength, and redirects the signal to another port. The switching process uses mirrors, reflecting the light rays (optical signals) to different ports and thus different cables or paths. Importantly, the OXC is reconfigurable, allowing dynamic adjustments to how signals are switched.

The OXC can also be customized. While its ports are usually equipped with multiplexers and de-multiplexers, some contexts may require certain ports to include transponders, adding a

connection to the electrical domain.

The OXC can perform additional functions, such as:



- **Grooming:** Aggregating multiple (typically low data-rate) optical signals using TDM into a single transponder. This transponder generates a new optical signal on one wavelength while retaining the distinction between tributary flows. Grooming maximizes transponder capacity, as transponders (being electrical components) have a finite data-rate capacity, unlike optical cables.
- **Wavelength Conversion:** If equipped with a transponder, the OXC can change a signal's wavelength.
- **Protection Against Failures:** By rerouting optical signals around points of failure.

Network Properties

Layered Functionality

The transport network, typically placed in the Data-Link layer (Layer 2) of the ISO/OSI Model, encompasses multiple nested layers of functionality. Within the optical transport network, functionalities akin to Network (routing), Data-Link, and Physical layers coexist, reflecting the complexity and layered structure of protocol stacks in the network.

Transparency Property

"Once the light path is set up, it can accommodate different types of service."

This means that, unlike in the electrical domain, light signals do not inherently distinguish between different services, enabling *Transparency*.

Advantages:

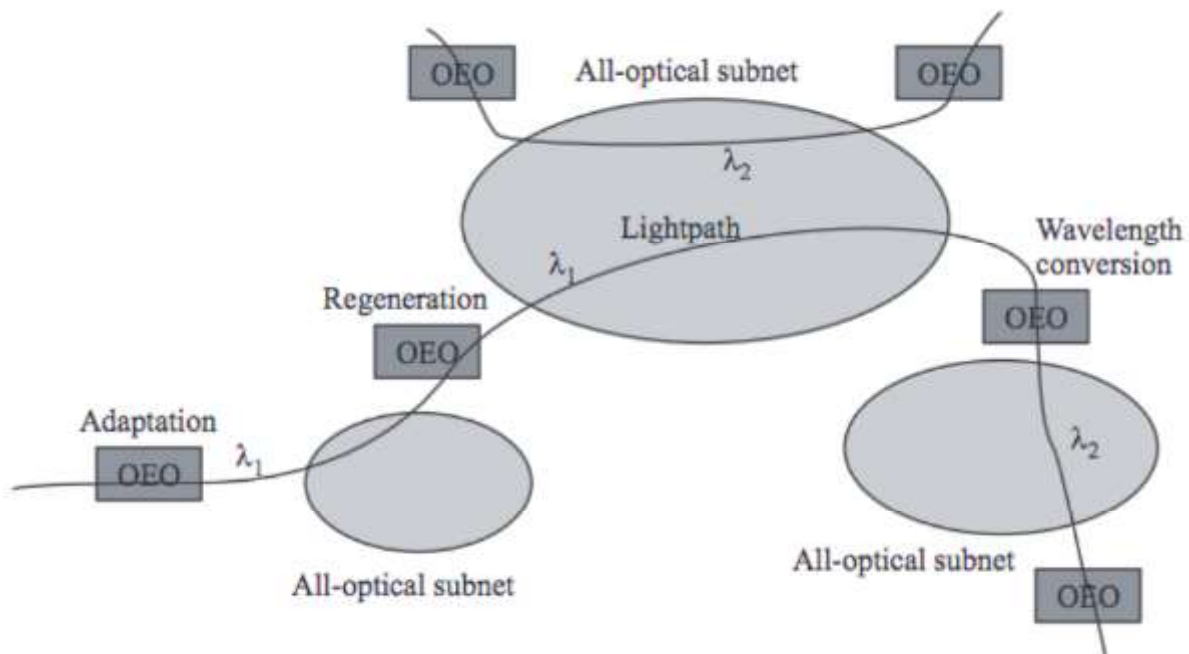
- Data is carried end-to-end in optical form.
- No optical-to-electrical conversions are required along the path.

Challenges:

- Analog signals require a higher SNR compared to digital signals.

Despite this, optical networks often incorporate electronics for intelligent control and management functions.

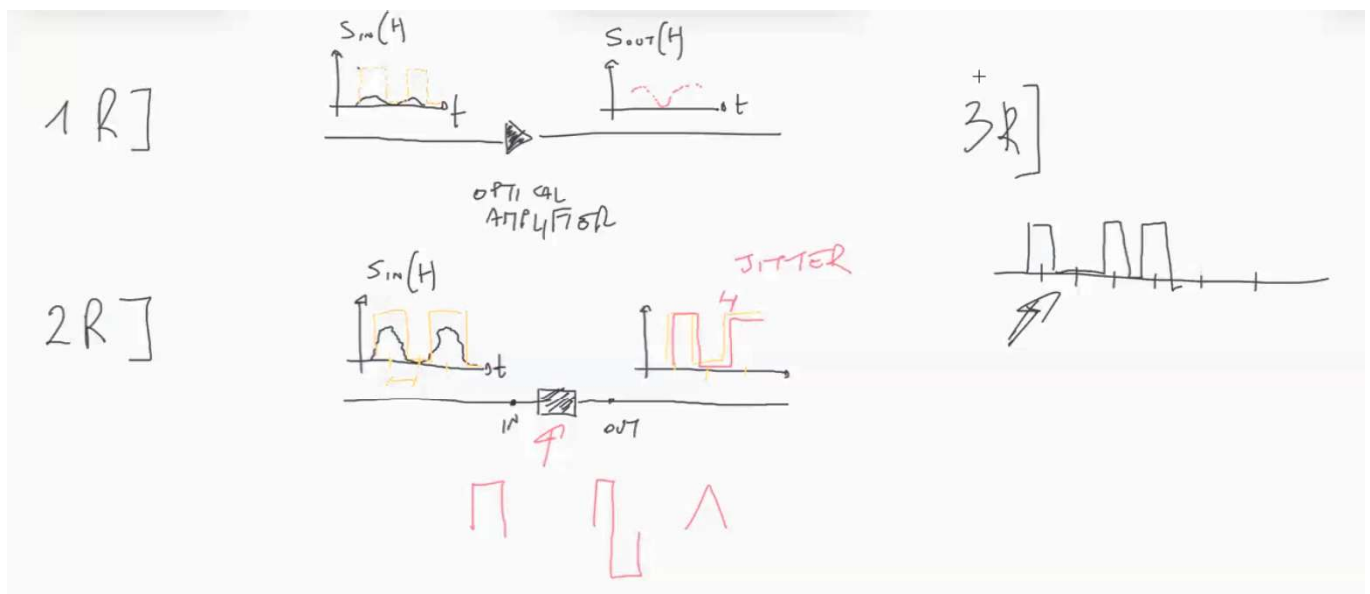
Sub-Networks and Domain Transitions



Transport networks are often divided into sub-networks, predominantly composed of optical components. Transitioning between sub-networks typically involves moving signals to the electrical domain using a transponder. This transition occurs in two critical network areas:

- **At the Network Edge:** To adapt client signals entering the optical domain.
- **In the Network Core:** For signal regeneration or wavelength conversion, addressing signal attenuation and degradation or unavailable wavelengths.

Signal Regeneration Types



Electronic regeneration restores signal quality and can be categorized as follows:

1. 1R: Regeneration

Optical amplifiers provide power to a signal, regenerating it entirely in the optical domain. While transparent, repeated use can distort the signal, making it an inefficient solution.

2. 2R: Regeneration with Reshaping

This process occurs in the electrical domain, assuming a signal's presence (light on = 1; light off = 0). It reshapes degraded signals using their original timing. However, it introduces two limitations:

- *Jitter*: Timing delays accumulate, potentially misrepresenting the signal.
- *Reduced Transparency*: Regenerators cannot distinguish signal types, requiring compatibility with specific impulse types and revealing potential technological details.

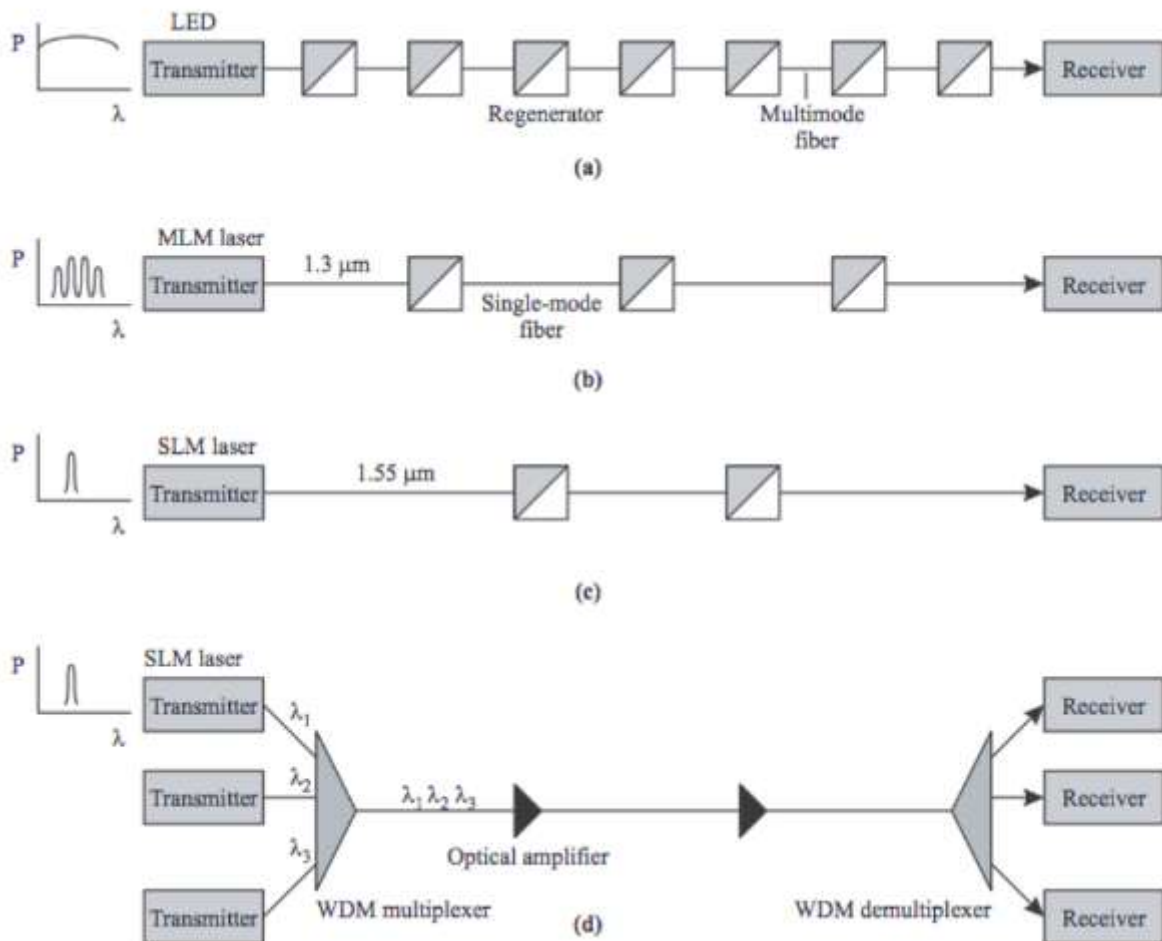
3. 3R: Regeneration with Re-Timing and Reshaping

This method recovers both the signal shape and timing, producing a signal closely resembling the original. While effective, it further reduces transparency, as it can reveal both the sender's data rate and technology.

Network Evolution

Note for Slides

When an element of the representation graphs present on the slide are grey, it means that element works on the electrical domain. Instead, when it's white, works in the optical domain. Elements that have both colors generally performs switches between the two domains.



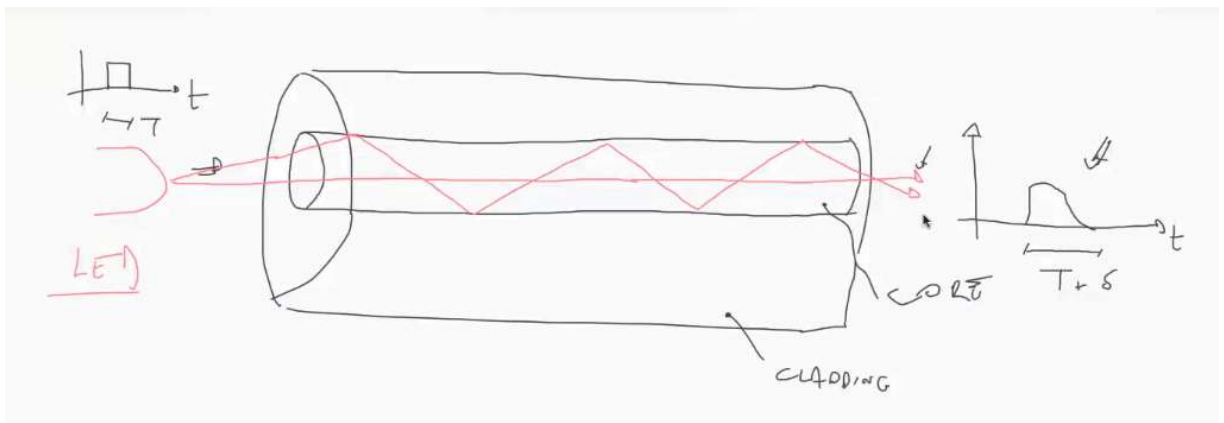
The figure above illustrates the evolution of optical networks and their transmission systems.

1. First Generation:

The graph λ (wavelength) $\rightarrow P$ (power) shows how the power spectrum is spread over a wide range of wavelengths. This indicates that the light ray (in this case, generated by an LED) emitted by the transmitter consists of contributions from different colors of light. Consequently, in this initial version of the network, WDM (Wavelength Division Multiplexing) was not feasible, as distinguishing users based on wavelength was impossible. Instead, TDM (Time Division Multiplexing) was employed as the multiplexing technique. The significant number of regenerators (3R in this case) further reflects how the network, at this stage, suffered from considerable transmission degradation caused by several phenomena:

- **Modal Dispersion:**

As depicted, the original optical fiber cables comprised two concentric glass cylinders: the **Core** and the **Cladding**. Light rays entering the Core traverse the fiber in different ways. Some take the shortest path, while others are reflected repeatedly by the glass walls. These varying trajectories lead to differences in travel time, causing signals to arrive later than expected. This temporal disparity distorts the signal's shape, degrading its quality.



This early network employed *Multi-Mode Fiber*, a type of fiber that permits the coexistence of multiple light rays, as shown in the example.

2. Second Generation:

This stage introduced several advancements:

1. The light ray was no longer generated by an LED but by a laser, transforming the $\lambda \rightarrow P$ graph into multiple impulses distributed across different wavelengths. The light ray remained multi-colored.
2. The operating wavelength was centered at $1.33 \mu m$, offering relatively flat attenuation and generally good performance.
3. Single-Mode Fiber was introduced. Unlike Multi-Mode Fiber, Single-Mode Fiber has a narrower core diameter. By reducing the possible trajectories of light rays, the modal dispersion was significantly decreased.

3. Third Generation:

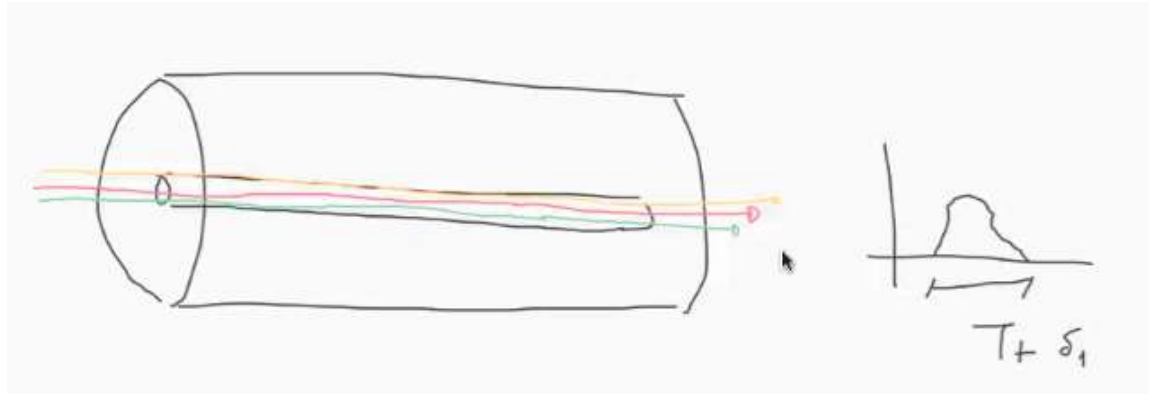
Key improvements in this stage included:

1. The laser emitted a light ray focused on a specific wavelength (TDM was still used). The wavelength was now centered at $1.55 \mu m$, where attenuation is much lower.
2. Transitioning from multi-mode to single-mode lasers helped mitigate another phenomenon:

- **Chromatic Dispersion:**

Chromatic dispersion occurs when light rays of different wavelengths (colors) travel through the optical fiber. Although they may traverse the same distance, their differing wavelengths affect their propagation speed. This leads to signal distortion, as some components arrive later than others. Chromatic dispersion

can be mitigated in two ways:



1. Using a more precise laser to produce a light ray composed of a single wavelength (*Note: The explanation provided in class for this phenomenon was unclear, as it conflated the problem with its solution*).
2. Increasing the density of the optical fiber core, thereby reducing timing differences caused by chromatic dispersion.

4. **Fourth Generation** (Current State):

The latest optical networks achieve maximum efficiency. Transmitters/transponders now generate light rays with focused impulses at a single wavelength. Multiplexing is performed using WDM, keeping operations within the optical layer as much as possible. This advancement also allows for the use of optical amplifiers, eliminating the need to transition to the electrical domain.

Control & Management

These terms correspond to the following functionalities:

1. **Control Plane** (or *Routing*): This function manages tasks such as establishing connections, defining light paths, and allocating suitable wavelengths.
2. **Configuration**: This involves configuring switching elements, assigning addresses, and setting up static routes. It also includes monitoring the network's health at regular intervals.

Initially, networks required manual configuration (typically via CLI). Over time, networks have become increasingly automated. Automation can be implemented in either a *centralized* or a *distributed* fashion:

- **Centralized Control:**

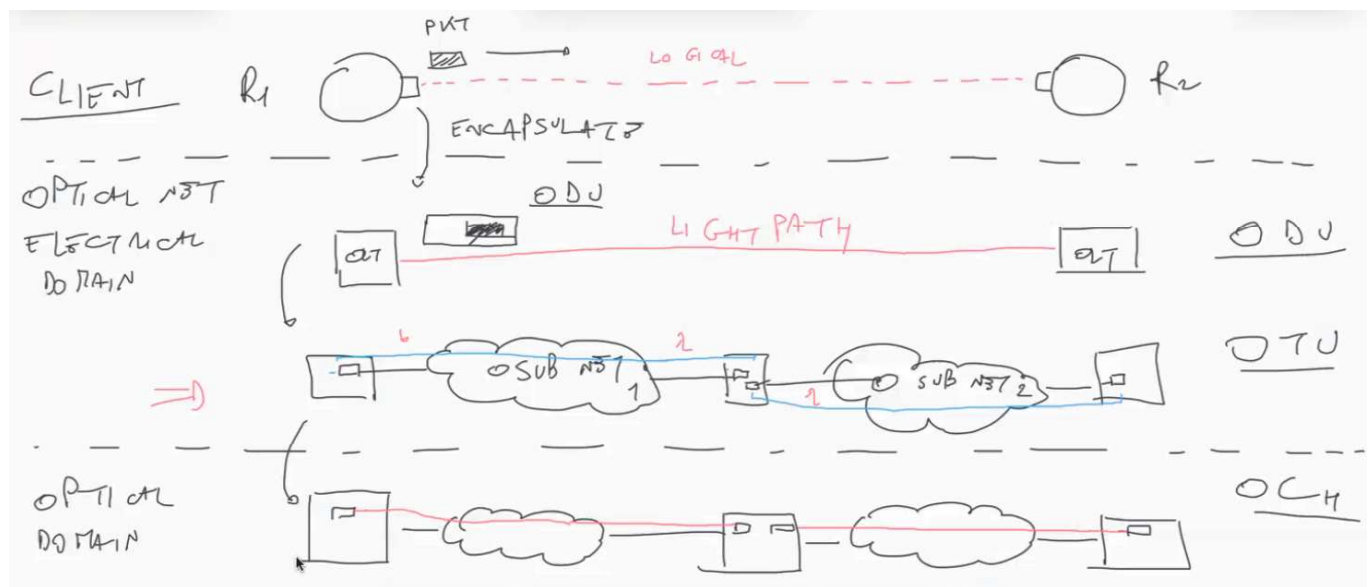
A central element, known as the *Control & Management Element*, centrally manages the optical network. For example, when configuring a new arrangement for an Optical Cross-Connect (OXC), the C&M Element executes an algorithm and communicates with the OXC via a specific protocol to adjust its switching metrics. When a client requests a new

connection, all operations are computed and initiated by the C&M Element. During light-path setup, the client layer can specify:

- The endpoints to interconnect.
- The required bandwidth.
- Whether an adaptation function is needed at ingress or egress points (e.g., for ITU-compatible wavelengths or transponder absence).
- The targeted Bit Error Rate (BER) threshold for sensitive traffic.
- The desired protection level against failures.
- Constraints on jitter and maximum end-to-end delay.

Multi-Layered Network

The optical network consists of a multi-layered infrastructure designed to handle various challenges such as bit-error checking, recovery, routing, and transmission. It includes the following layers:



1. Client Layer:

This abstract layer provides the logical connection between clients. Although clients transmit packets, these are always encapsulated before being sent to lower layers. Encapsulation adds overhead to ensure the packets function in subsequent layers.

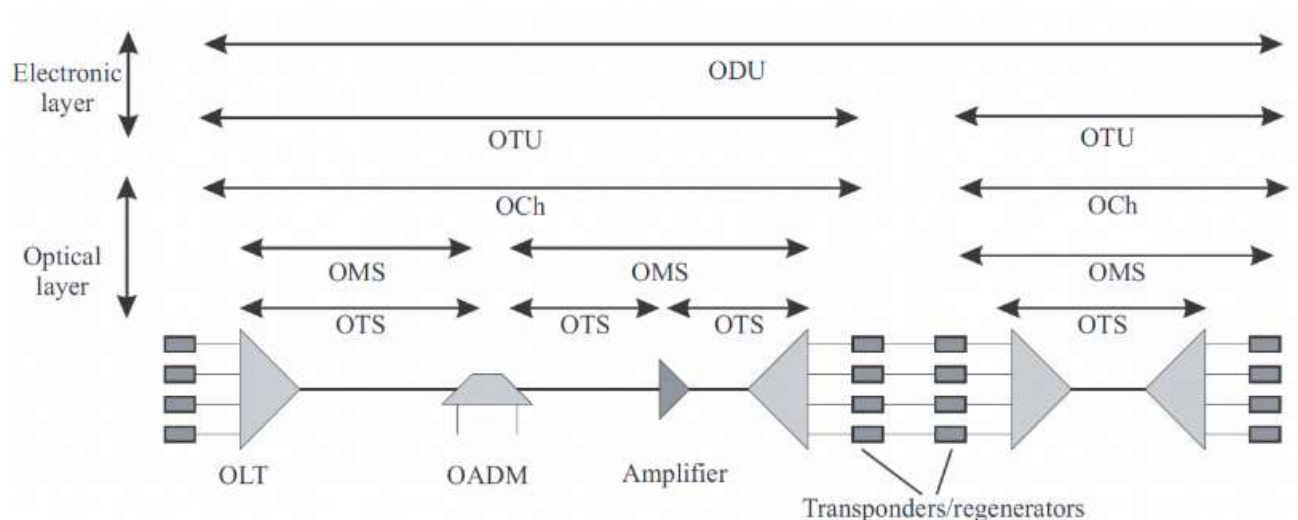
2. Electrical Domain:

- **ODU Layer:** This layer features two OLTs at either end of the connection, represented as a single, straightforward light path. Packets, encapsulated with additional overhead, become Optical Data Units (ODUs). This layer primarily interacts with the client layer.

- **OTU Layer:** Moving deeper, we gain a clearer understanding of system transmission. What appears to be a simple light path often traverses multiple optical sub-networks. At the edges of each sub-network, OLTs play a crucial role, particularly their transponder components, which enable wavelength conversion. Wavelength uniformity is maintained within each sub-network, making wavelength conversion frequent at sub-network boundaries.

3. Optical Domain:

- **Optical Channel (Och):** This layer represents the portion of a light path between two transponders. The complete light path consists of concatenated optical channels. Transponders and other devices at this layer interact with the OTU Layer to perform their functions, including tracking the number of sub-networks a signal traverses (involving transitions back to the electrical domain).
- **Optical Multiplexing Section (OMS):** This layer delves into optical sub-networks, identifying elements like OLTs and Optical Add-Drop Multiplexers (OADMs).
- **Optical Transmission Section (OTS):** This is the most granular layer, detailing individual fiber sections between devices, regardless of multiplexing. It reflects the actual physical topology of the optical network.



Performance and Fault Management

To provide guaranteed quality of service to end users, constant monitoring of both performance and fault management is essential. This process involves the following:

Performance Management:

1. Monitoring performance parameters for all connections.
2. Taking necessary actions to ensure desired performance goals are met.

Fault Management:

1. Detecting problems in the network.
2. Alerting management systems appropriately through alarms.
3. Attempting to recover lost services in case of a fault.

Bit Error Rate (BER) and Optical Trace

BER:

- The **Bit Error Rate (BER)** is a critical performance metric for lightpaths.
- BER detection is only possible when the signal is in the electrical domain, typically at regenerator or transponder locations.
- Overhead inserted in OTN (Optical Transport Network) frames, consisting of parity check bytes, enables BER computation.

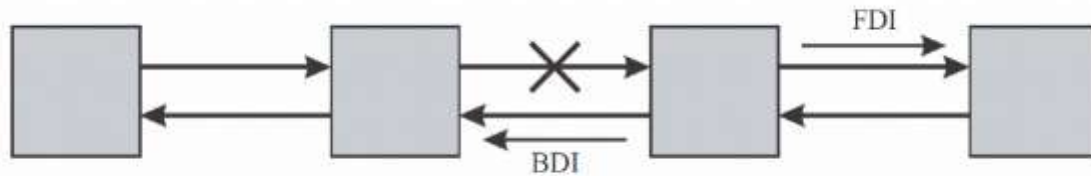
Optical Trace:

- Lightpaths traverse multiple nodes and multiple cards within the equipment at each node.
- A **unique identifier**, called the **optical path trace**, is associated with each lightpath.
- This trace helps the management system identify, verify, and manage lightpath connectivity.
- The trace contains at least the following four values:
 1. ID of the client sender.
 2. ID of the client receiver.
 3. ID of the transponder on the left side.
 4. ID of the transponder on the right side.

Alarm Management

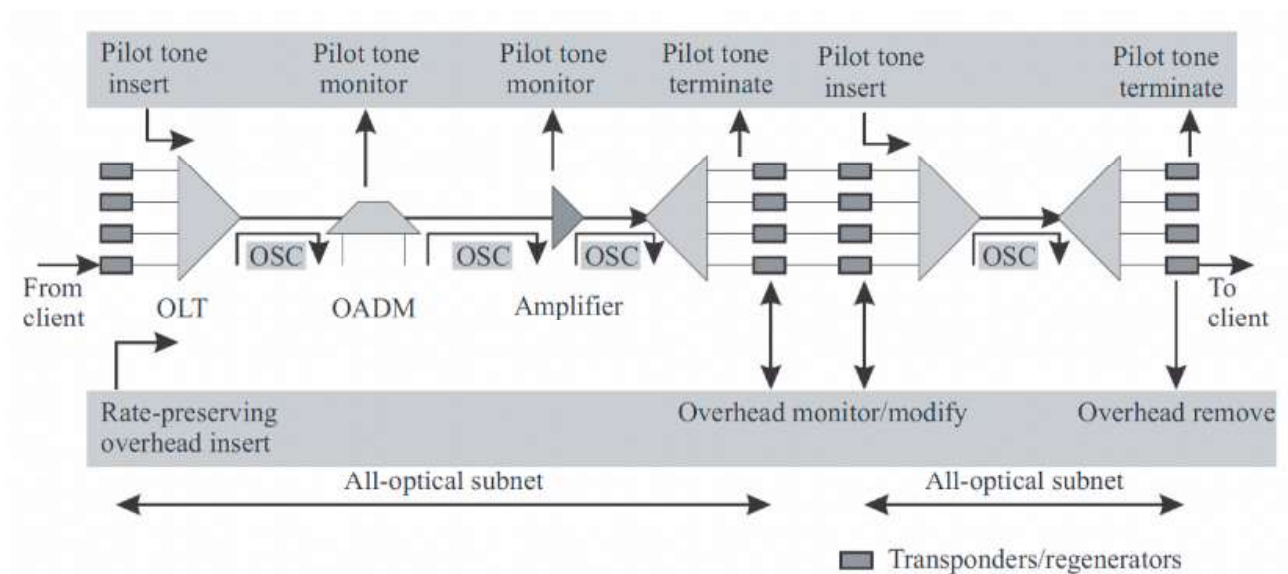
In optical networks, a single failure event can generate multiple alarms. For example, in a network with 32 lightpaths on a given link, each traversing two intermediate nodes, the failure of a single link could trigger a total of 129 alarms.

Alarm Suppression:



- Alarm management identifies the root cause of the failure and suppresses redundant alarms.
- This is accomplished using special signals:
 - **Forward Defect Indicator (FDI)**: Sent downstream to the next node to notify them of the failure and suppress alarms further downstream.
 - **Backward Defect Indicator (BDI)**: Sent upstream to notify the previous node of the failure.

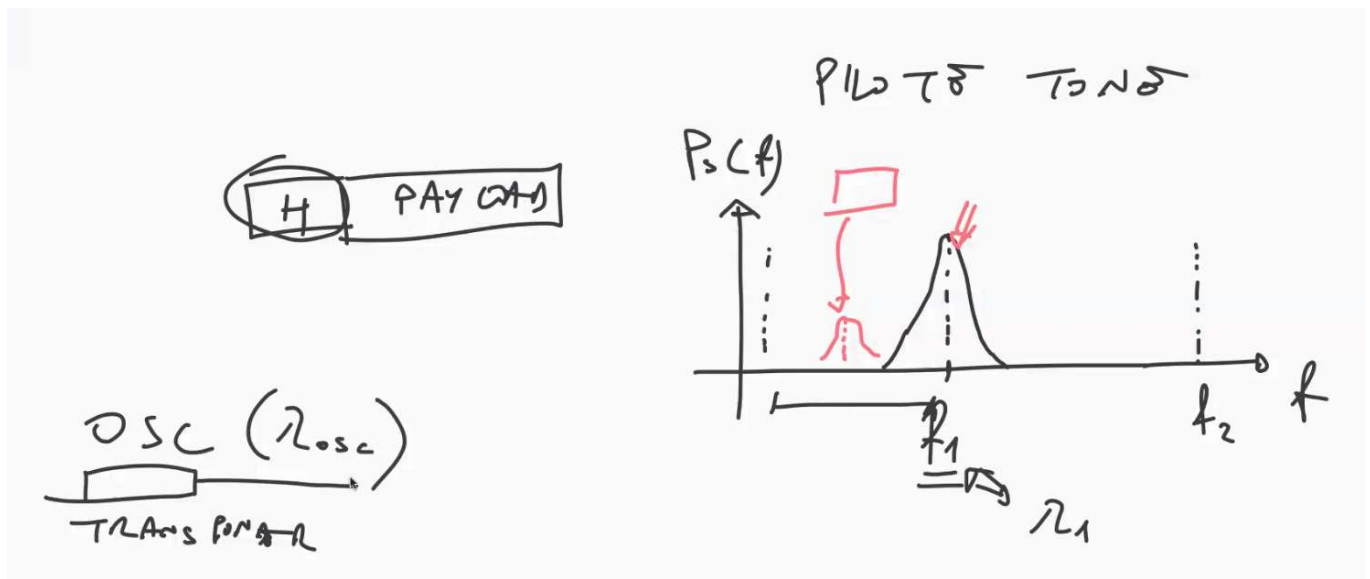
Operation:



- FDI and BDI signals are sent at different sub-layers of the optical layer.
- A node receiving an FDI or BDI stops sending alarms. This ensures that only one node, closest to the fault location, continuously raises alarms.
- Devices sensing network issues continue to raise alarms until they receive an FDI or BDI.
- FDI and BDI are transmitted multiple times across different layers to ensure all devices are informed of the fault's location and nature.

OSC and Pilot Tone

Protocols and functions in the optical network, such as BER, path trace, and defect indicators, require special overhead. Two methods are used to achieve this:



Pilot Tone:

- Any wavelength is centered at a certain frequency, with gaps between signals for distinction.
- The **Pilot Tone** is placed in one of these gaps, utilizing low power and data rate.
- Messages, encoded in binary, are modulated and converted into optical signals sent via the Pilot Tone band.
- The Pilot Tone is added as overhead to the **Optical Channel Layer (OCh)** by the transponder and is terminated at the end of the OCh.
- It reflects faults or errors in the associated client signal.

Optical Supervisory Channel (OSC):

- The **OSC** uses a reserved wavelength for control-monitoring purposes, independent of client signals.
- It has a single-fiber lifespan and monitors fiber health.
- Unlike the Pilot Tone, the OSC is not client-signal-specific and can be applied anywhere in the network.

Applications of OSC and Pilot Tone

Application	All-Optical Subnet		End-to-End
	OSC	Pilot Tone	Rate-Preserving
Trace	OTS	OCh	OTU ODU
DIs	OTS OMS OCh	None	OTU ODU
Performance monitoring	None	Optical power	BER
Client signal compatibility	Any	Any	Any

1. Trace:

- The OSC can trace multiple client signals within the **OTS (Optical Transmission Section)**.
- Since an OTS section may carry signals from different clients with individual traces in the Optical Channel, the OSC wavelength can provide additional trace overhead.

2. Defect Indicators:

- The OSC covers defect indicators across all optical layers (OTS, OMS, OCh).
- Devices without a transponder for generating a Pilot Tone rely on the OSC for fault communication.

3. Performance Monitoring:

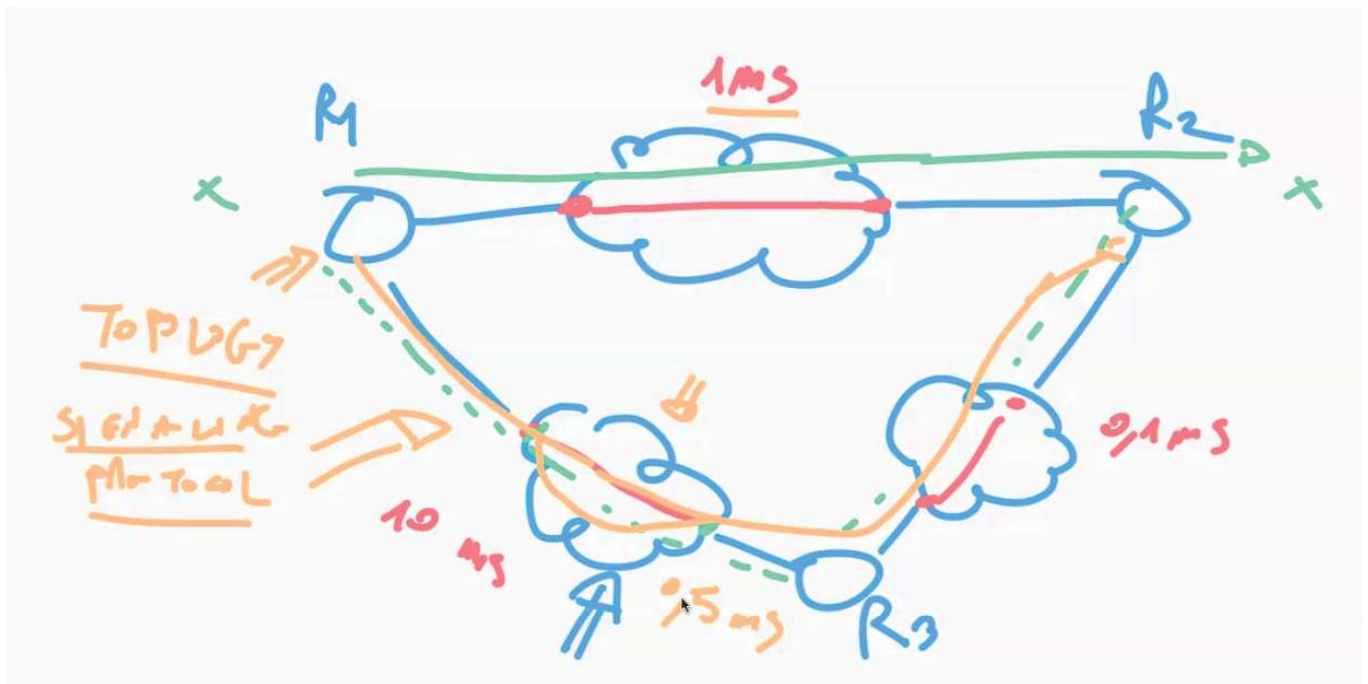
- In the optical domain, the Pilot Tone monitors the optical power of client signals. If the Pilot Tone experiences attenuation, the client signal is likely affected similarly.
- In the electrical domain, BER techniques evaluate signal degradation.

4. Rate-Preserving Overhead:

- Overhead carried in Optical Network headers without reducing client data rate is referred to as **Rate-Preserving Overhead**.

Client-Server Model

In the client-server model, the client layer (e.g., IP, SDH, ATM) requests the server layer (the optical network) to establish a connection (light-path) without having knowledge of the internal structure of the optical network.



The server layer includes the **MCE (Management & Control Element)**, which, when asked to create a new connection, executes optimization algorithms to arrange the needed resources and determine the light-path from the client to the destination. The optimization algorithm takes as input:

- The topology of the optical network
- The status of the network (a complete list of available wavelengths on every single link)
- Quality of Service (QoS) parameters specified by the customer, such as jitter, BER, and reliability

The output of this function is a suitable light-path as requested by the client. This model is highly transparent and can accommodate a wide variety of customers. However, the light-path created in this model is definitive and does not change during the communication.

- **Centralized control:** The setup and management of light-paths are controlled centrally.
- **Best suited for infrequent light-path setups:** This model is ideal when connections do not need to be established or altered rapidly.

Peer Model

The peer model assumes that the owner of the client network also owns the provider network, enabling control-plane functionalities across both networks. In this setup, the **CPF (Control Plane Function)** of each router at both ends of the connection collaborates with the MCE to determine a suitable light-path for the end-user.

A key distinction of this model is that the light-path can change dynamically during communication. If the CPF detects a more optimized path to an intermediate point or the

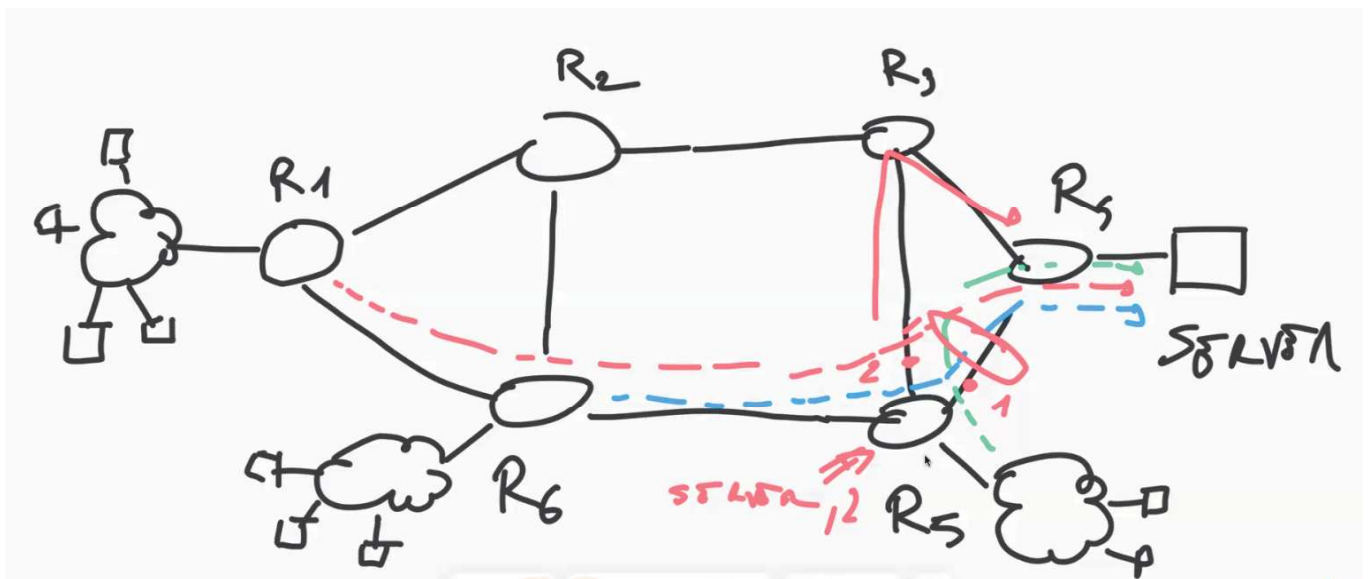
destination, it can reroute the connection accordingly.

For a router to operate in this model, it must have access to:

- **Network topology:** This is obtained via OSPF (Open Shortest Path First). Routers continuously receive topology updates from devices along the network.
- **Signaling protocols:** When the router detects a better path, it reserves the necessary resources, such as wavelengths, to establish the new route.
- **Tight coupling between client and optical layers:** The optical layer primarily serves a single client, such as IP.
- **Distributed control:** This model is managed in a distributed manner across the network.
- **Best suited for frequent, dynamic connection setups:** This model is advantageous when connections need to be established, altered, or taken down rapidly.

MPLS

Before MPLS

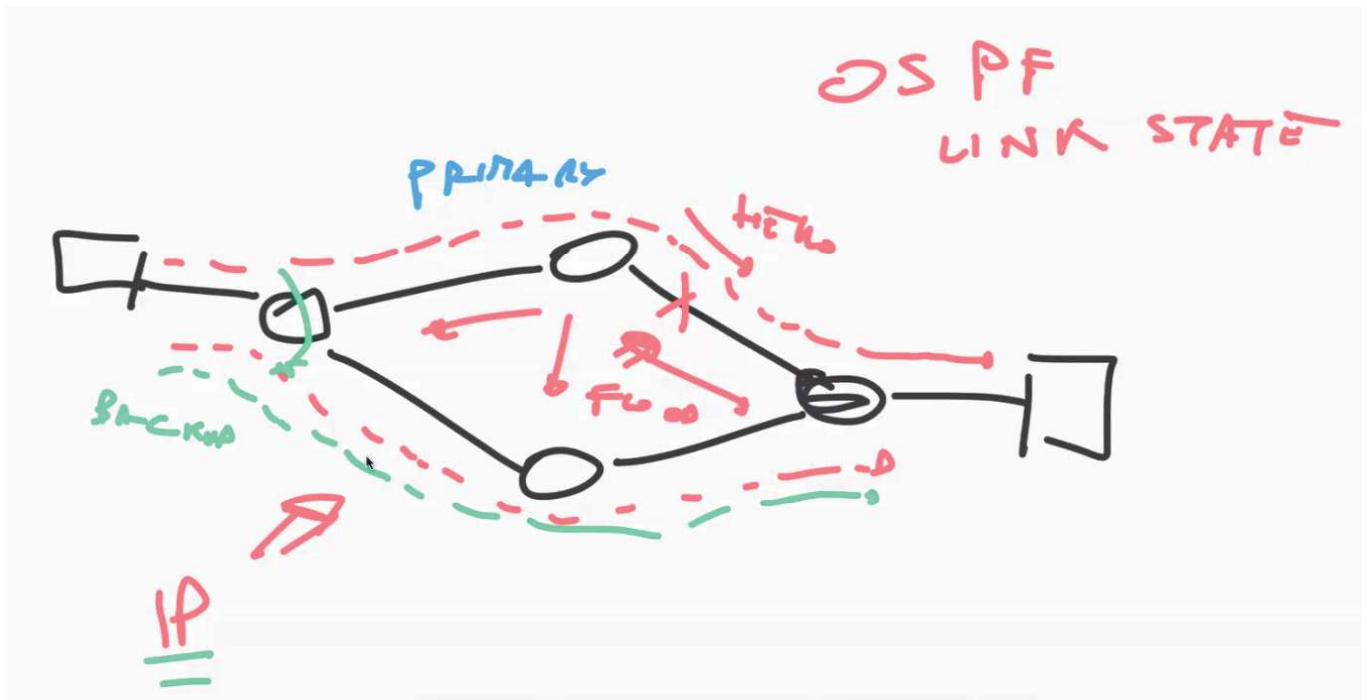


The starting problem is the possibility of congestion, multiple data flows that exceed the maximum capacity chosen, occurring in one section of the network. Rerouting all of the traffic does not resolve the issue since we are only shifting the problem from one location to another. There are actually two main solutions to this:

- **Traffic Engineering:** Fairly distribute the traffic across all links in the network, aiming to balance the data flows and avoid the congestion that initiated the problem in the first place.

- **VPN (Virtual Private Network)**

There are two ways to handle failures in an IP network:



- Through the **OSPF (Open Shortest Path First) Protocol** (a Link-State Protocol). This is done by continuously sending *Hello* messages to neighboring nodes, and if a node stops sending them, it indicates a problem has occurred. Once a failure is detected, all routers recompute the new shortest path and redirect traffic accordingly.
- Through a **Back-up Link**, which is pre-configured as an alternative path. If a link in a path fails, the traffic is rerouted to the backup. However, it is not possible to create this backup dynamically during communication or after network configuration.

Both of these methods have limitations, which led to the invention of a new protocol designed to handle rerouting dynamically at the IP layer: the **Multi Protocol Label Switching (MPLS) Protocol**.

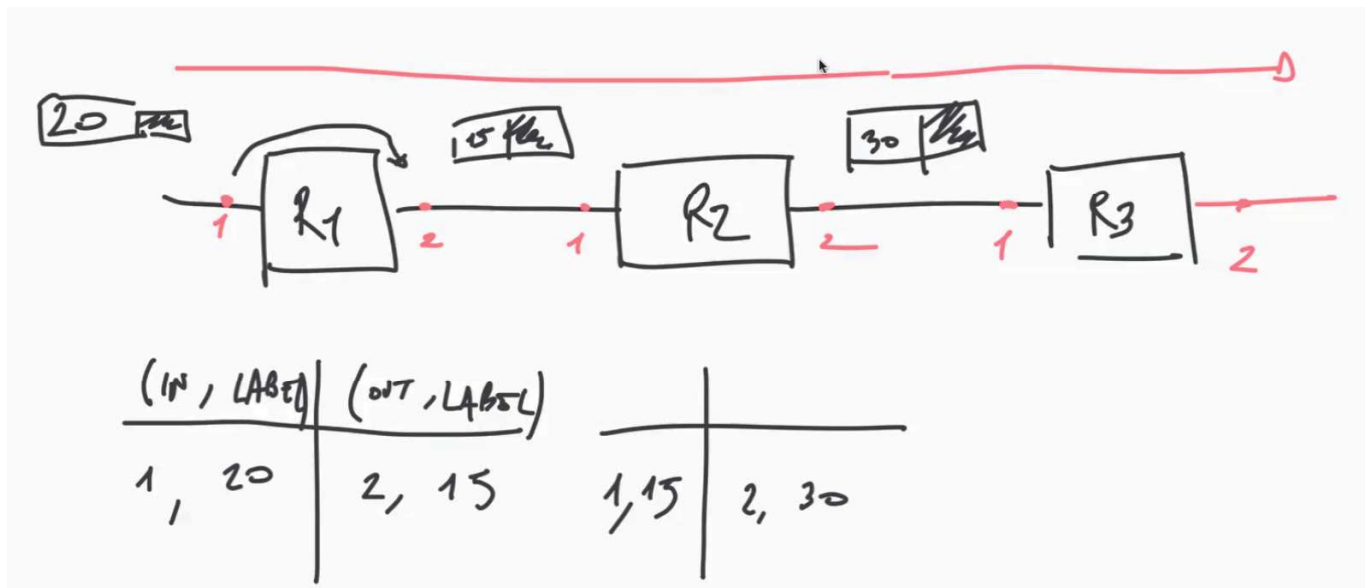
MPLS Protocol

MPLS is a technology that enhances the routing services provided by the IP Layer. In particular, it turns the IP Layer from a connectionless environment into a connection-oriented one.

Initially, the POS Standard (IP over SDH/SONET) was the foundational standard under the IP Layer, constituting the first generation of optical networks (e.g., TDM). The second generation includes the MPLS Protocol, which extends the services provided by the IP Layer and

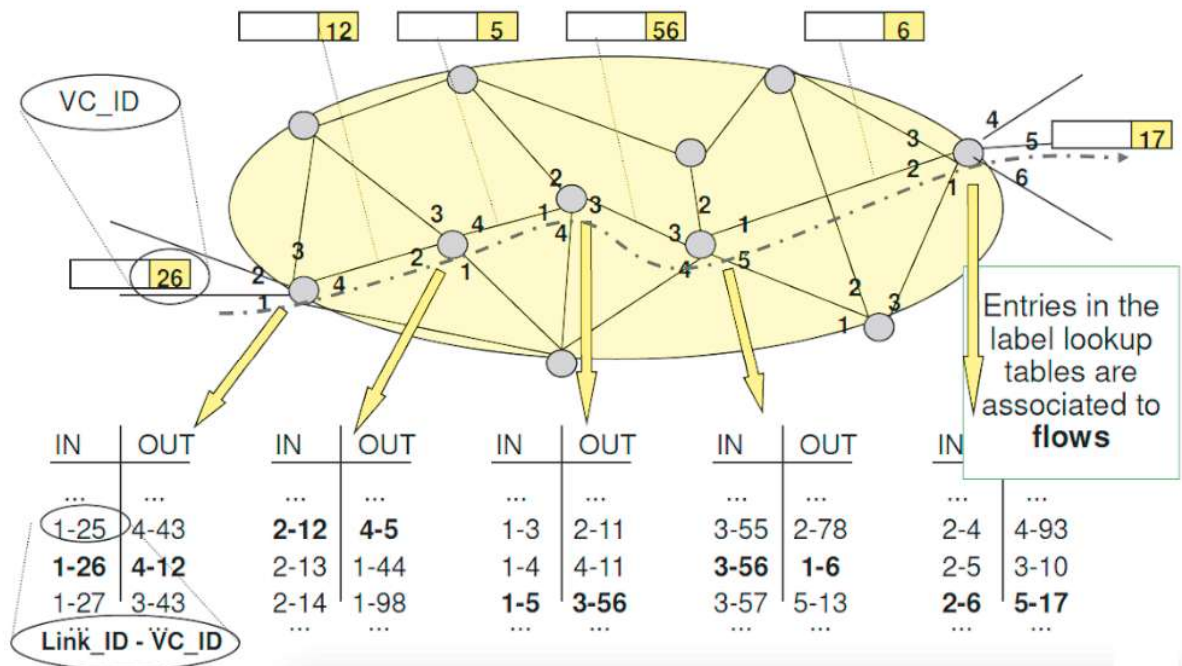
encompasses functionalities such as WDM, which we are studying in this course. Generally, MPLS is referred to as **Layer 2.5** (between Layer 2 and Layer 3).

The **Multi-Protocol** aspect of MPLS refers to its ability to work with any Layer 2 and Layer 3 protocol types, providing significant flexibility. The **Label Switching** part refers to the practical functioning of the protocol.



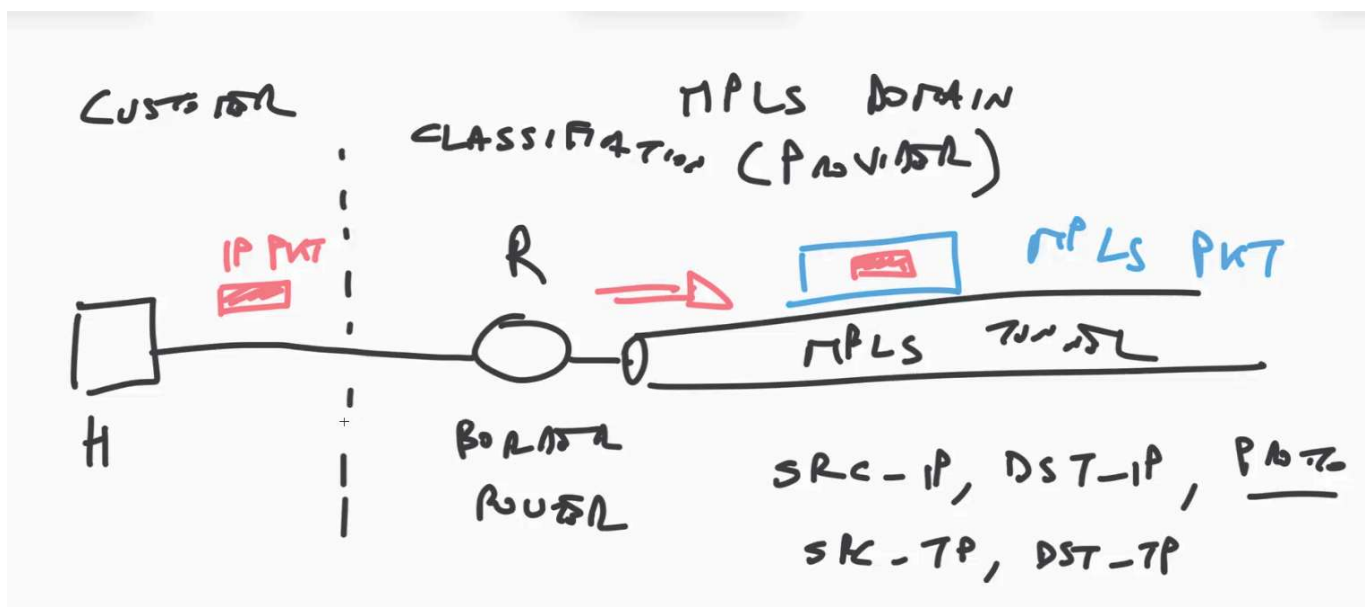
Unlike the IP Protocol, which performs routing based only on the destination, the MPLS Protocol performs routing based on the concept of a *label*. MPLS is a connection-oriented paradigm, as entries must be made in the routing tables of each node. An entry consists of two fields:

- The **IN (Incoming)** field, which refers to incoming packets.
- The **OUT (Outgoing)** field, which refers to outgoing packets.



Each field contains two pieces of information: the **label** and the **port**. The label (an integer) is attached to the packet and changes every time the packet passes through a switch. The new label value is determined by the **OUT** field of the routing table. The Port field specifies the port of entry (IN field) and the port of exit (OUT field). Therefore, the label has only local validity.

A single node may route two different packets with the same label value but through different ports, which does not lead to any conflicts. However, MPLS cannot support two different packets with the same label value that are directed to the same local destination (same node, same port) or the same link. This is considered a mistake in MPLS.



MPLS establishes a series of predetermined paths for packet routing, resembling traffic on a railway or through a tunnel (Traffic Steering). This *tunneling* effect, present in other protocols

like SSH, involves **encapsulating** the packet and sending it over a predetermined path. Determining which MPLS tunnel to use for a specific packet is called **classification**. The router classifies based on five fields from the original IP packet:

- IP address of the source
- IP address of the destination
- Protocol (IP, TCP, etc.) — located in the IP header
- Source transport port
- Destination transport port

Using these fields, the router identifies the most suitable path and reserves the necessary resources to establish the MPLS tunnel. Creating such tunnels is a distributed process, supported by the family of protocols known as **Label Distribution Protocols**. Some classification rules must be manually configured in routers to establish these criteria.

A key consequence of this process is the possibility of having multiple MPLS tunnels running concurrently, even starting and ending at the same nodes. This is made possible for two main reasons:

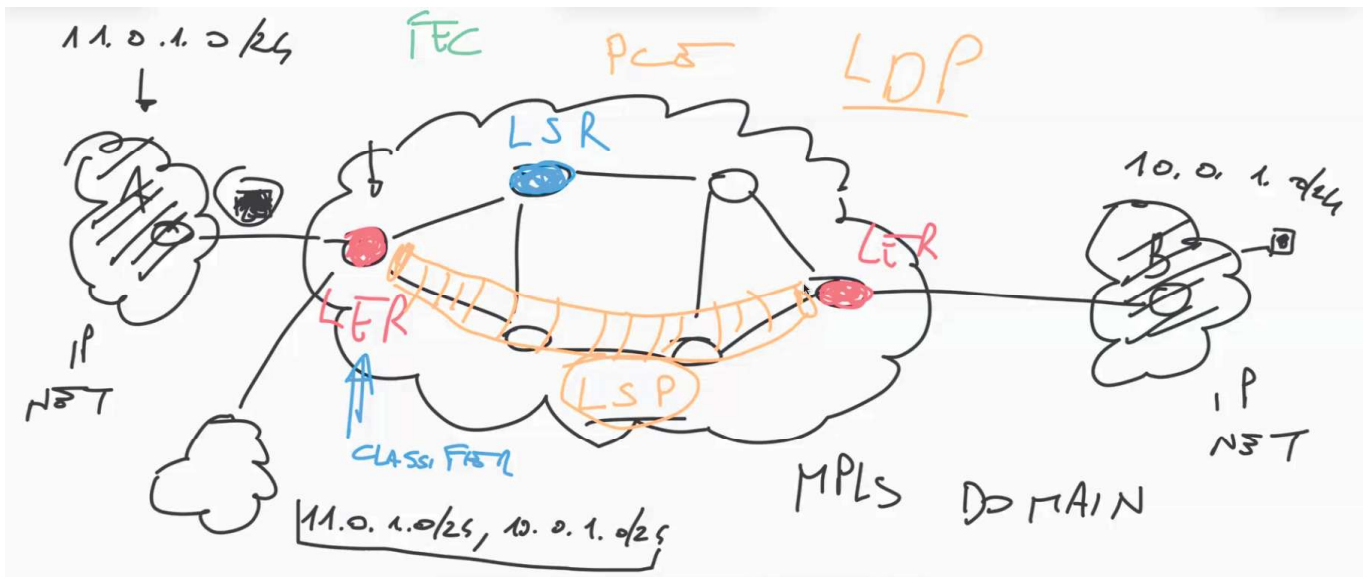
- **Classification**: Packets can be classified to use different tunnels even if they start and end at the same points. The rationale for classification typically involves distinguishing between different hosts.
- **Label**: A further step involves assigning different labels to packets at the starting and ending nodes.

Another property of MPLS routing is that each sub-path is always the shortest path to its respective source and destination nodes (and links). This is beneficial for distributed protocols but results in the IP Control Plane directing a significant amount of traffic through a limited set of shortest paths, as these links are considered optimal from a topological perspective.

Types of Services Improved by MPLS

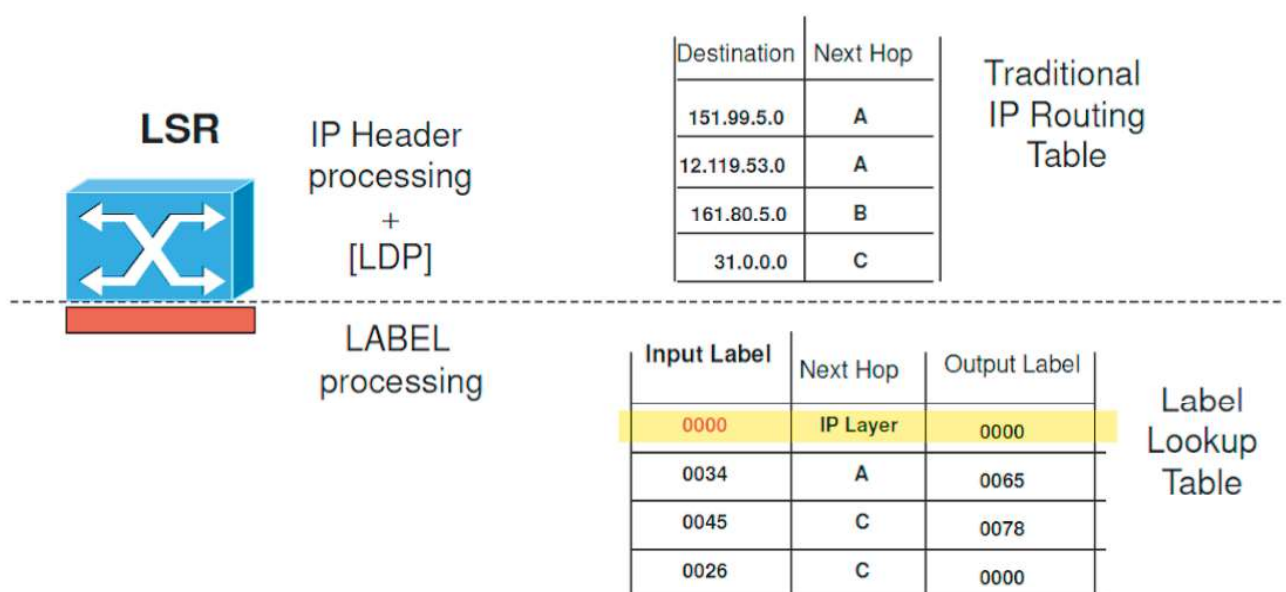
- **VPN**
- **Traffic Engineering**
- **Fast Recovery**

Any OLT (Optical Line Terminal) that supports the IP Protocol also supports MPLS. The **Control Component** and the **Forwarding Component** are integrated within an IP router, which becomes an “IP/MPLS router.” The MPLS Forwarding Component acts as an extension of the IP forwarding capability.



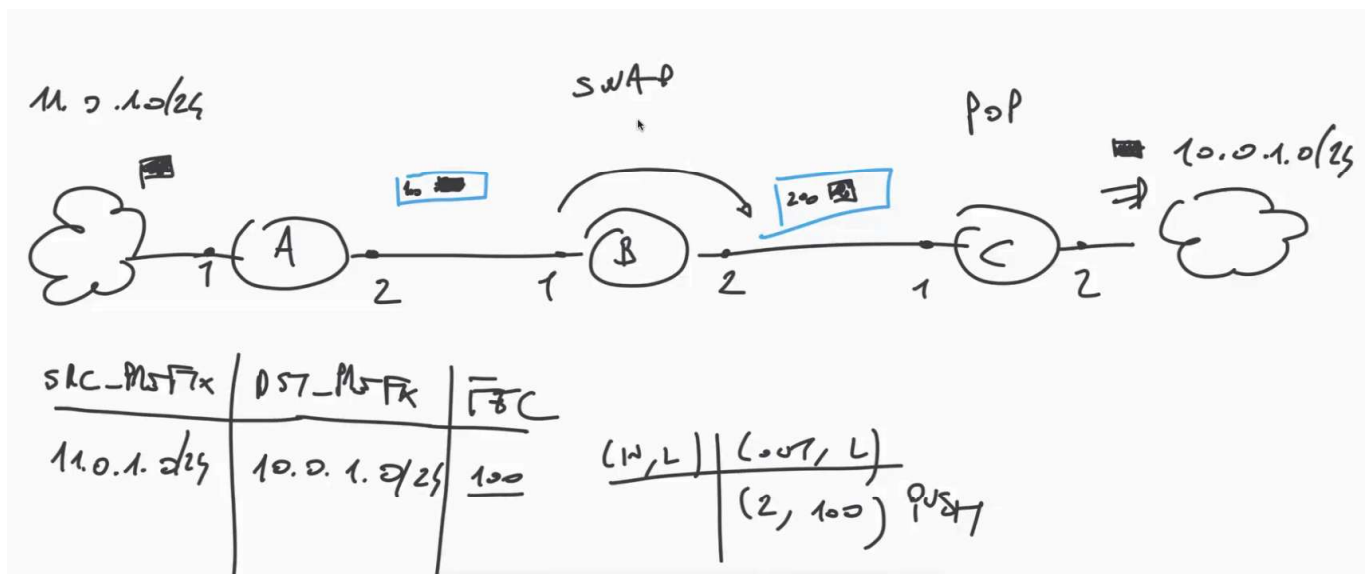
The MPLS label is carried in an MPLS header (also known as a “shim header”) inserted between the Layer 2 and Layer 3 headers (Layer 2.5). The header is 32 bits long:

- **Label** – 20 bits
- **Exp** – 3 bits: (*We don't care*)
- **S** – 1 bit: If this flag is set to 1, the next header is the IP header; otherwise, it is another MPLS header.
- **TTL** – 8 bits: Time to Live field
- **Label Edge Router (LER)**: A border node for an MPLS domain that forwards packets to and from the MPLS domain, inserting and removing labels for packets entering and leaving the domain.
- **Label Switching Router (LSR)**: A node that performs label switching within the MPLS domain.



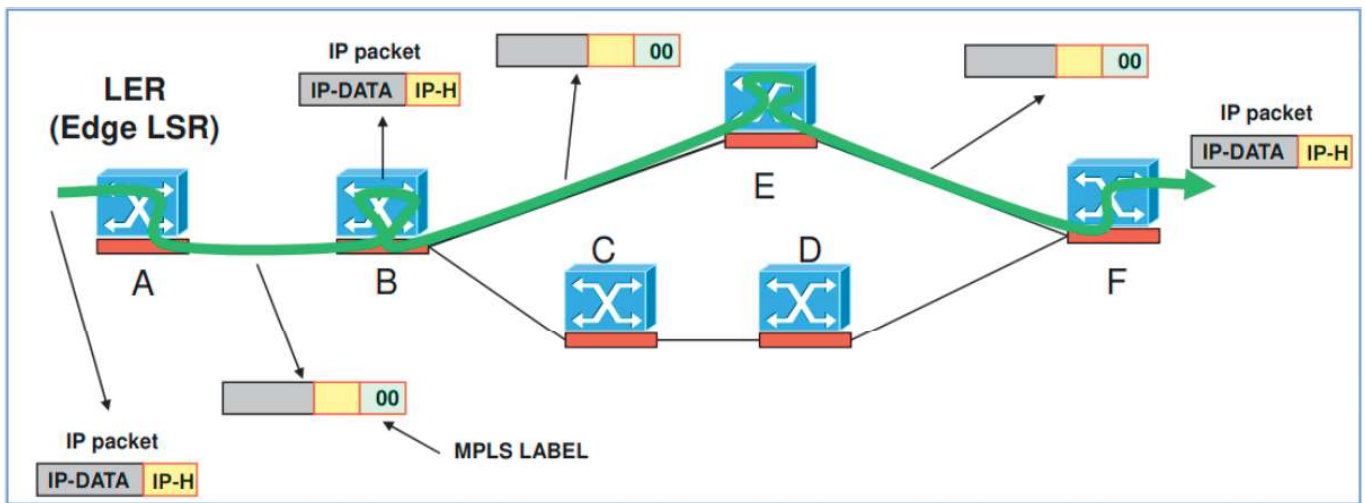
- **Label Distribution Protocol (LDP):** Alongside traditional IP routing protocols, LDP is used to distribute labels among MPLS devices.
- **Forwarding Equivalence Class (FEC):** A set of IP packets that are forwarded in the same way and receive the same treatment.
- **Label Switched Path (LSP):** The path across one or more LSRs followed by packets belonging to a FEC, corresponding to the “Virtual Circuit” concept.
- **Ingress LER:** The MPLS domain analyzes the IP header of the packet, classifies it, adds the MPLS label, and forwards it to the next-hop LSR.
- **In the MPLS domain:** The packet is forwarded along the LSP according to the labels.
- **Egress LER:** Removes the label, and the packet is forwarded based on the IP destination address.

Three basic actions:

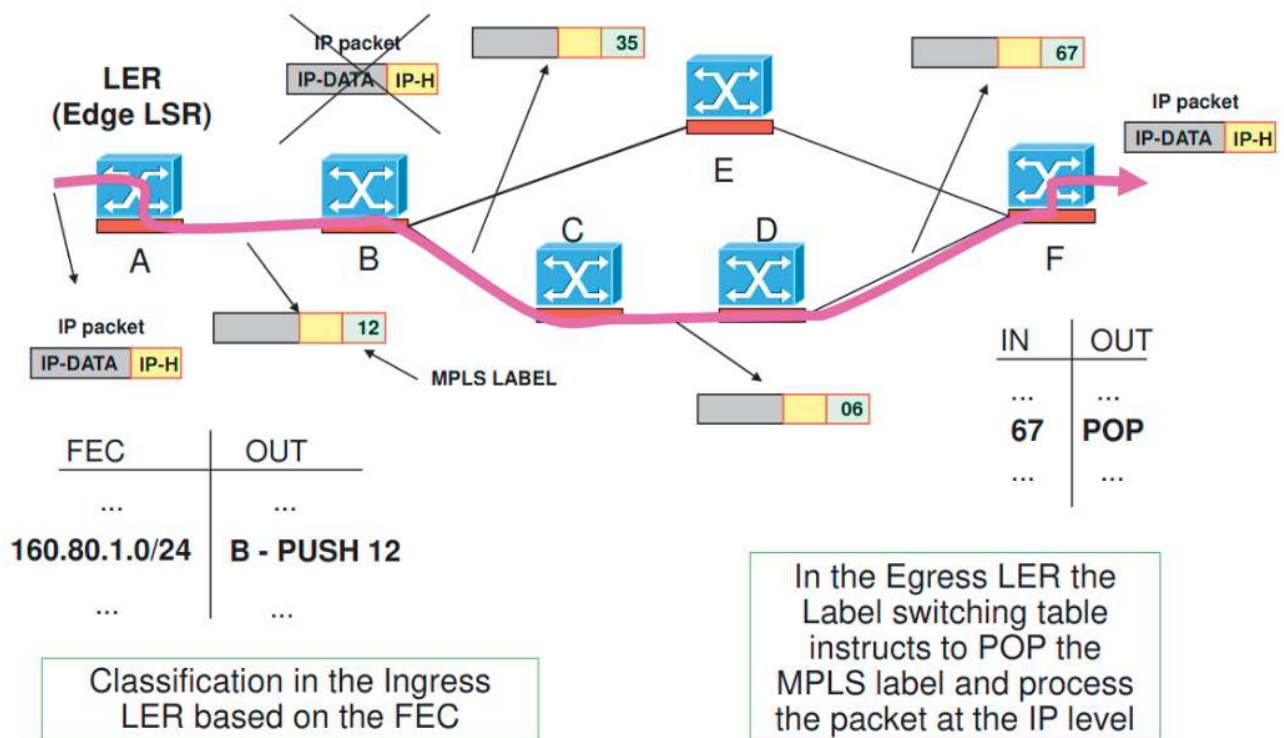


- **Push operation** – Add the MPLS label.
- **Pop operation** – Remove the MPLS label.
- **Swap operation** – Change the label (performed at an intermediate node).

Based on the quality of service requirements or other factors, there are two possible ways to forward IP packets:

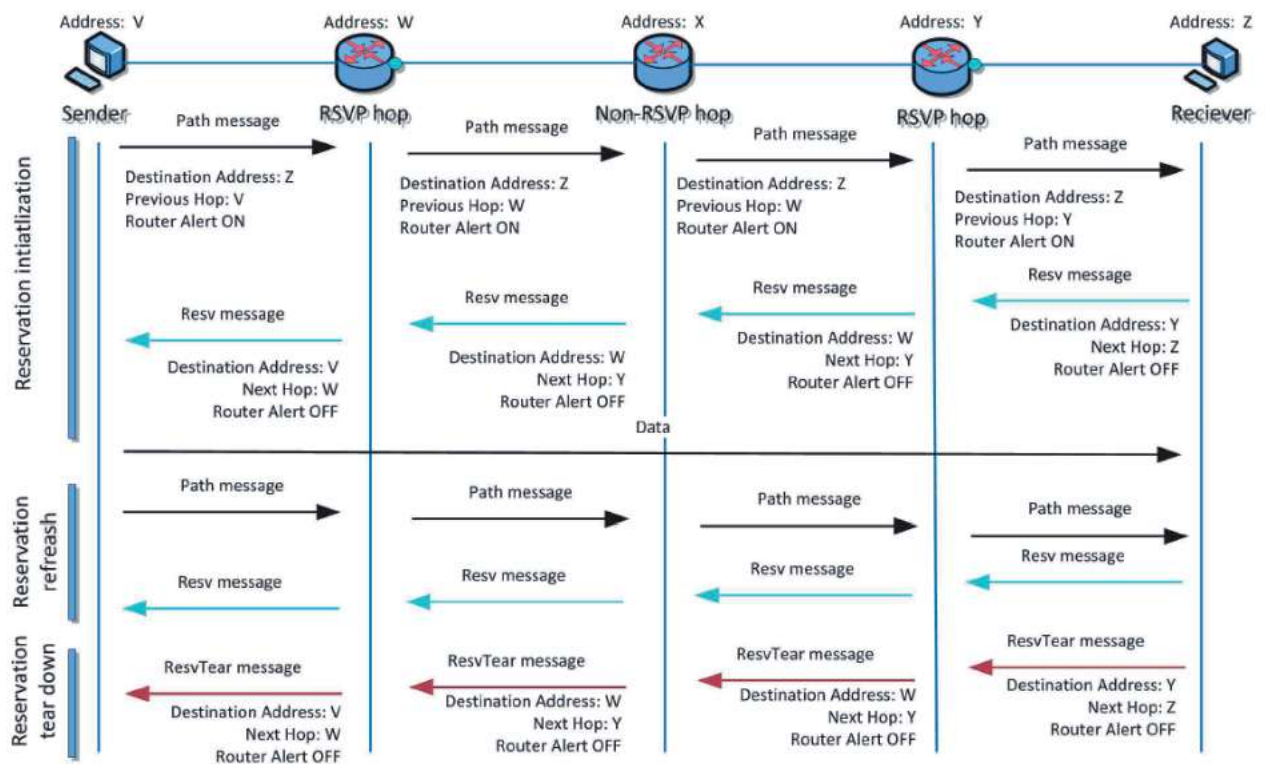


- **Connectionless forwarding along the IP path:** This ensures the packet follows the shortest possible path to the destination, using Best-Effort/ IP Protocol. This option is enabled by using the 00 label in the MPLS header.



- **Connection-oriented forwarding along an LSP:** In this option, the LER classifies the IP packet based on its FEC and forwards it into an LSP tunnel.

How to Set Up an LSP

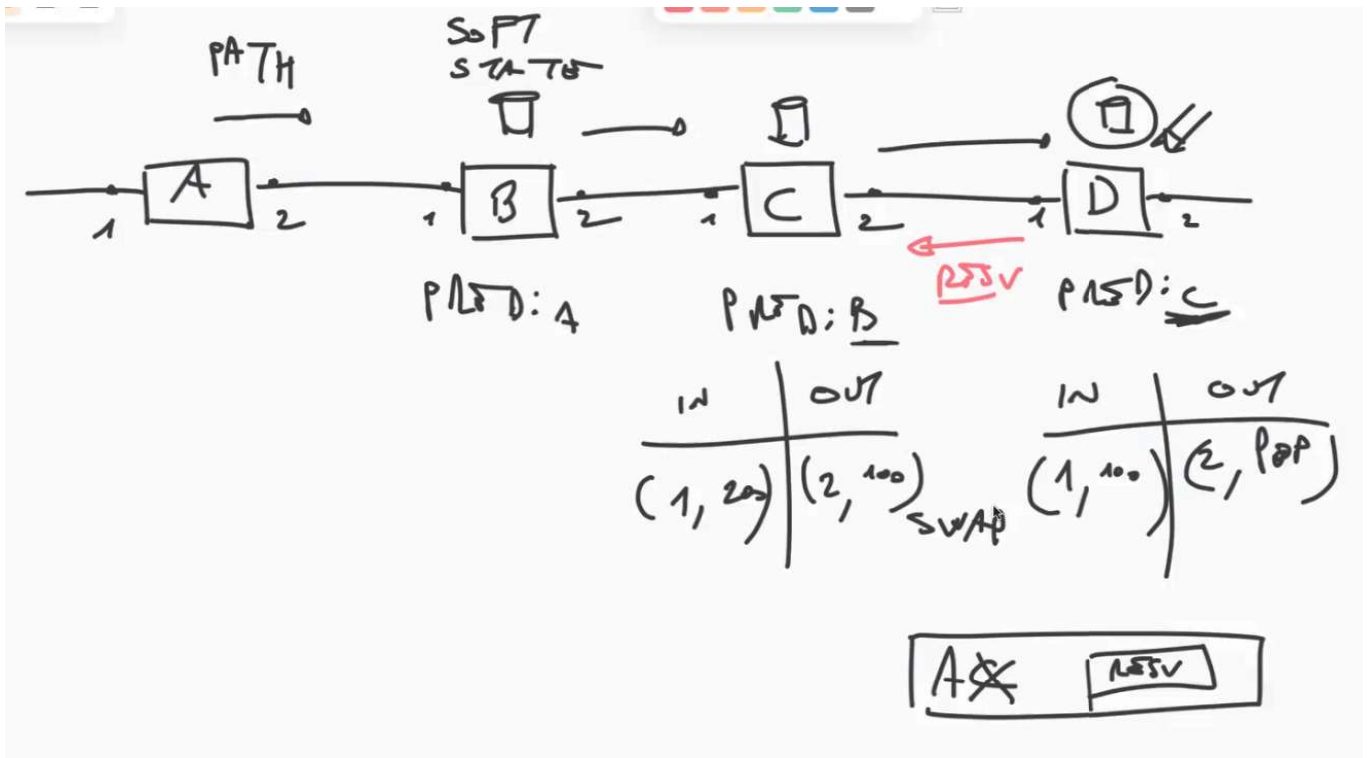


So, when we want to create an LSP, these are the steps:

1. Request from the client
2. Net looks for a path with given bandwidth etc. (not necessarily the shortest)
3. instruct the switches to how to deal with the packets

For doing point 3, we can use a simple *Label Distribution Protocol*:

The LER sends a special type of message called a *Path* message to other LSRs. If the desired path is the shortest, the *Path* message can be encapsulated within an IP packet. Otherwise, we can specify only the next-hop in the network. The *Path* packet travels across the chosen path (*What a cool name, huh?*).

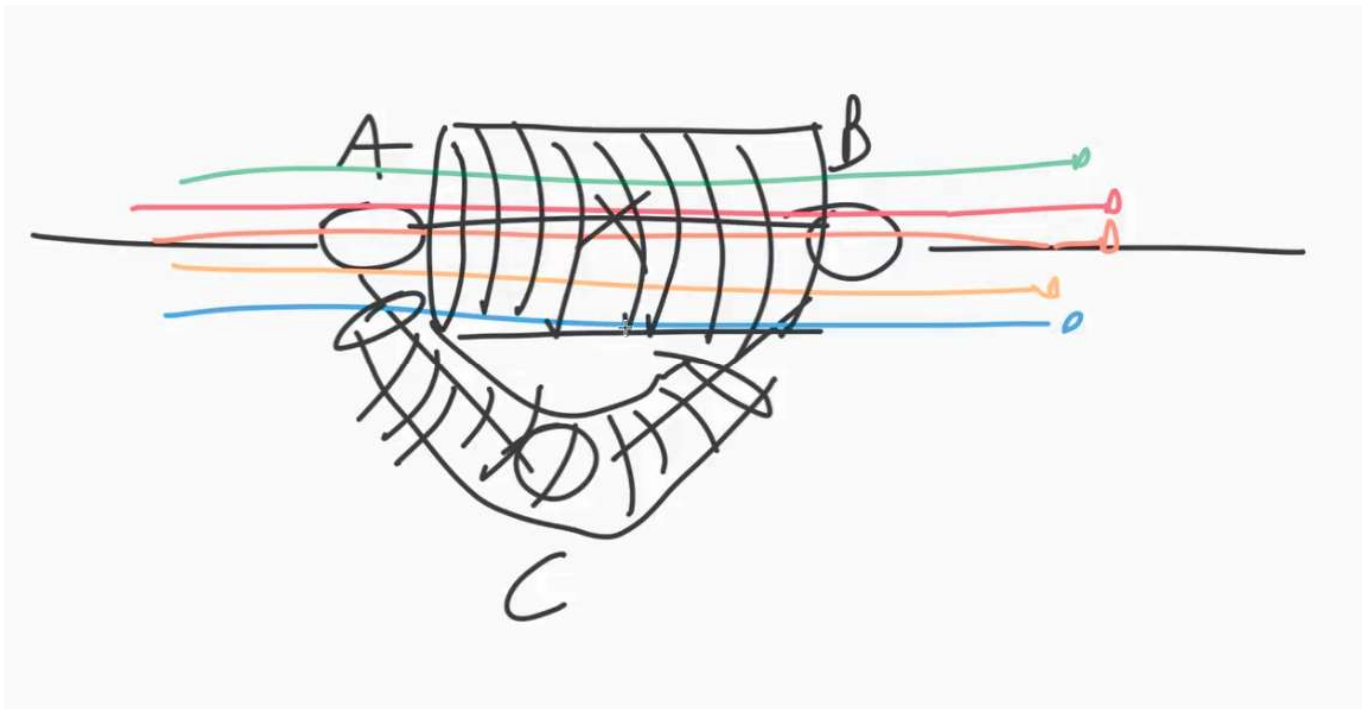


Every node maintains a *soft-state*, a variable that stores the predecessor of the original Path message. When a node receives a *Path* message, it remembers the last sender of the message. Once the *Path* message reaches the destination LER, this node sends back a *Resv* message in the opposite direction, specifying to the predecessor nodes (in the soft-state) which **label** to use on the corresponding links between them.

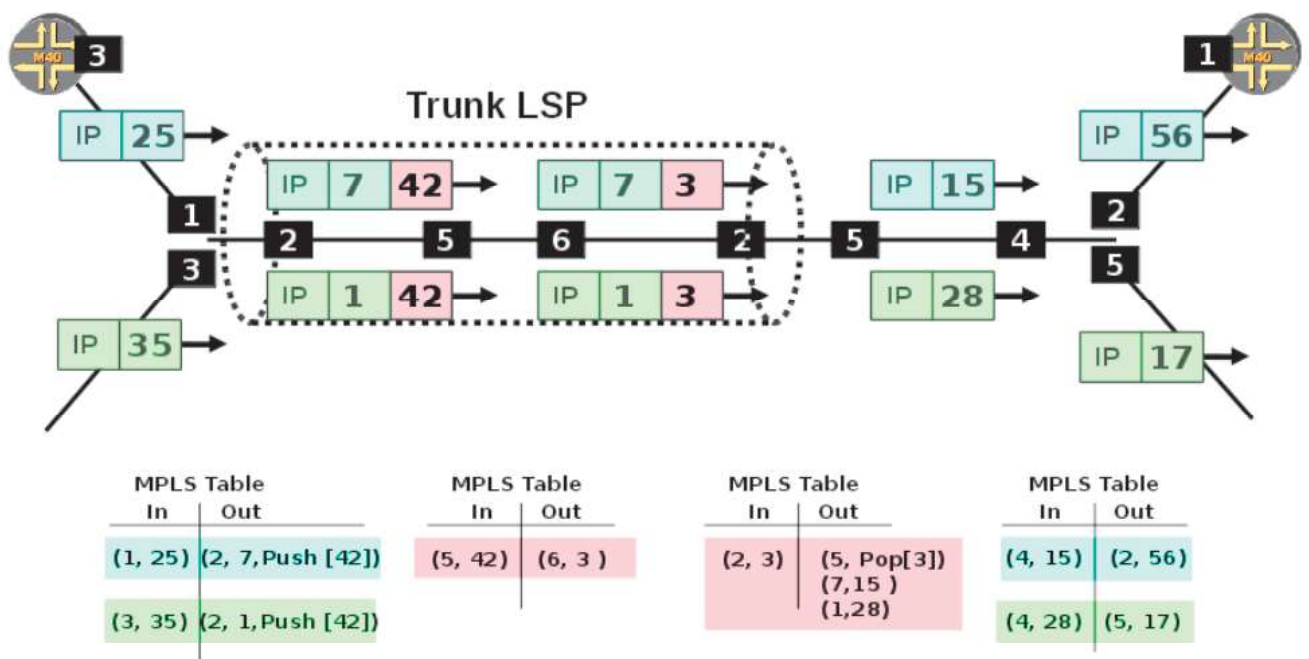
The soft-state serves another purpose: since two nodes may be linked in both directions, **and the link in one direction might differ from the other direction**, the soft-state allows the *Resv* message to travel along the same path as the *Path* message. This entire process is the **Connection Initialization**, which is always present in connection-oriented protocols.

Once the MPLS connection is established, the *Path-Resv* mechanism is periodically performed to reconfirm and re-establish paths after detecting failures.

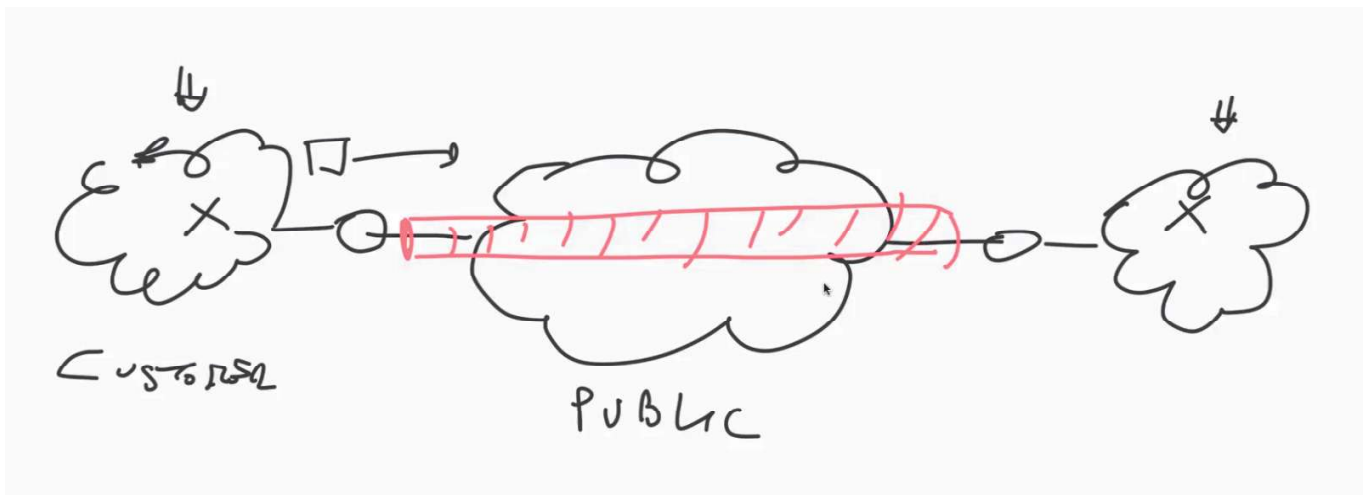
Label Stack



When a node performs the **Push operation** on an MPLS packet, it adds an additional MPLS label. This technique, called Label Stack, is used for more efficient control of multiple data flows. In other words, the additional label acts as a **nested MPLS network**, which simplifies packet rerouting.

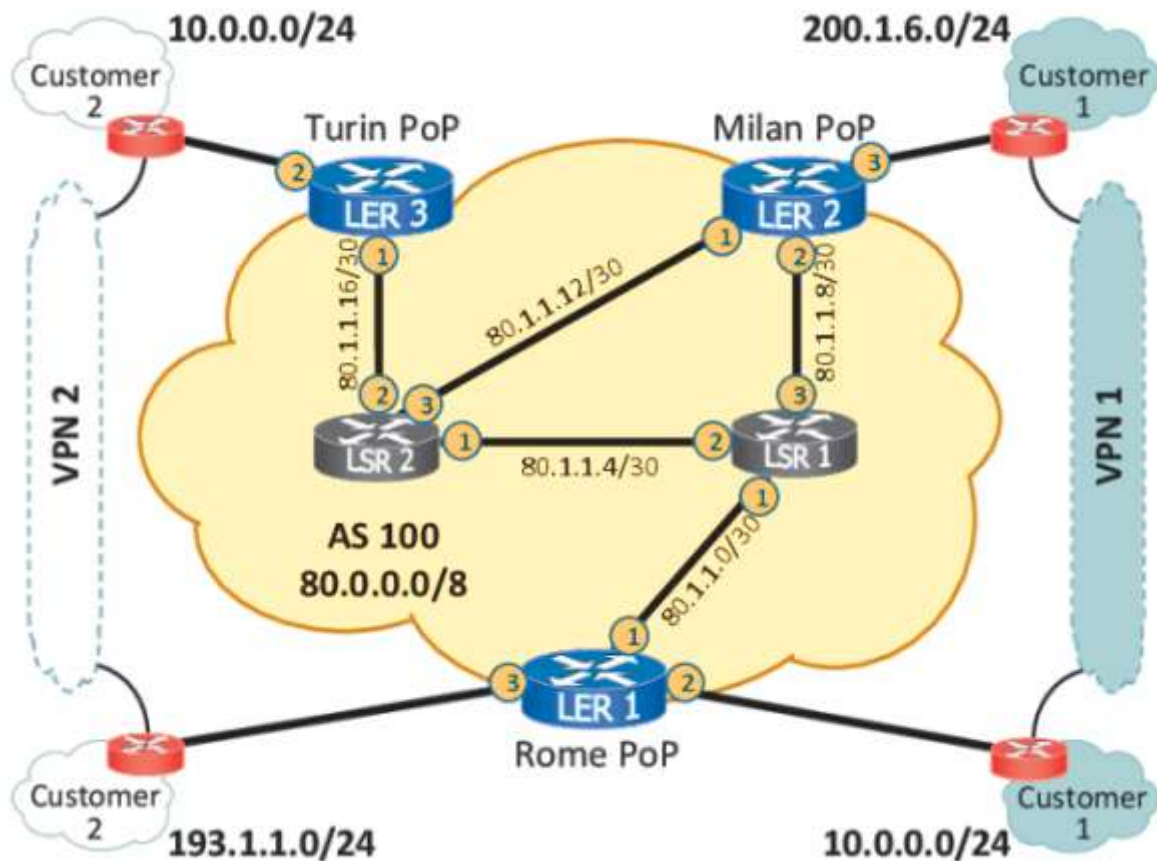


VPN Service



The main features of VPN are **security/privacy** and **support of private IP addressing**. Let's consider two private networks: one in Rome and the other in Milan, and we want to connect them. We have two options:

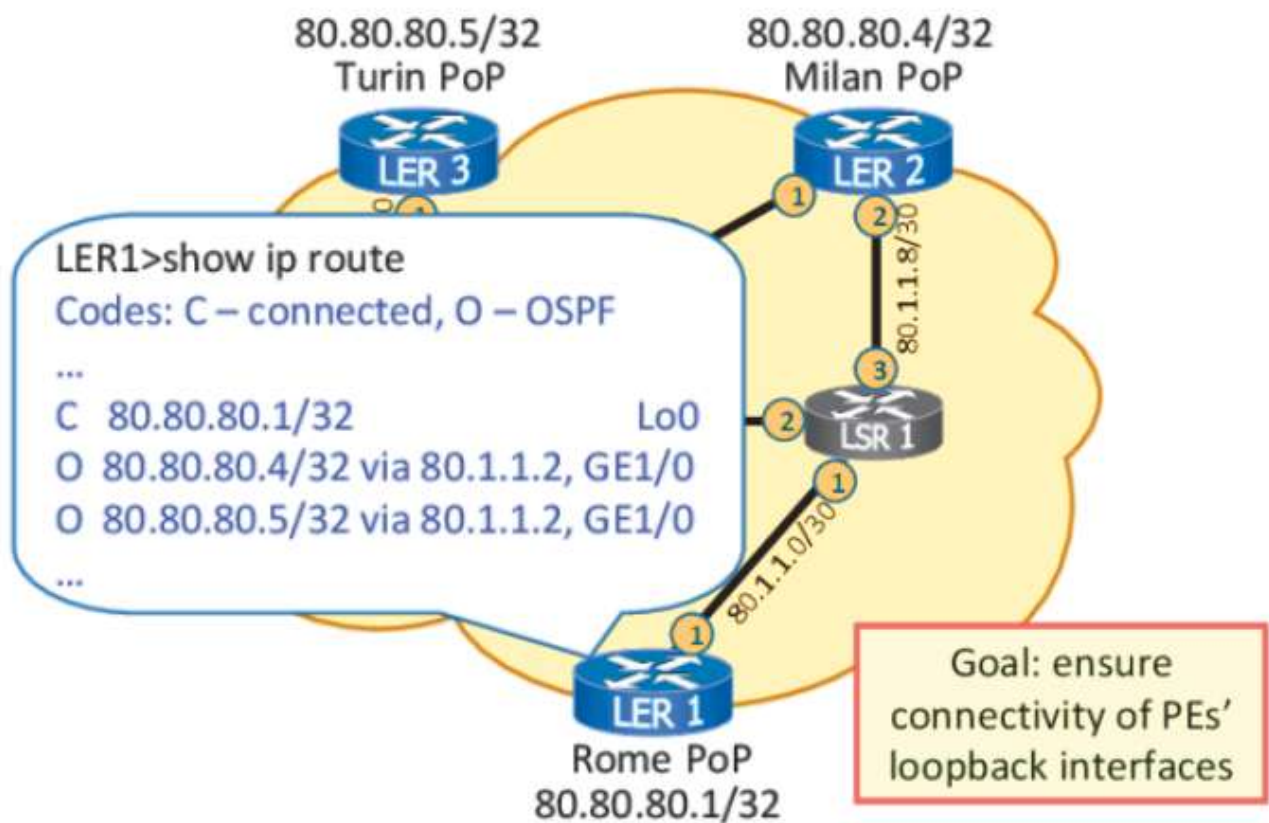
- **Option a:** Leased Line – a dedicated private line. This is unrealistic and expensive.
- **Option b:** Use the public network (Internet), creating a tunnel where only the two ends communicate. To achieve this, we will need a **VPN (Virtual Private Network)**, since the two networks use private address spaces that are not routable externally (in the Internet). A VPN encapsulates the packet in another packet, with the outer packet having a public IP that is routable on the Internet.



This forms a **TN (Tunnel Network)** that covers a wide area. There are two customers: Customer 1 has a network in Turin, and Customer 2 wants to connect its networks in Turin and Rome. Can we use a NAT protocol on a GW (Gateway) for this? Let's see: No, we can't, because in our case both source IP and destination IP are private addresses. We need a VPN. A VPN is a tunnel realized over a public network, through which packets are sent. The service provider has multiple customers, each wanting to create its VPN. All the devices belonging to the MPLS network are called **PE (Provider Edge)** routers, while the devices owned by the customer are **CE (Customer Edge)** routers.

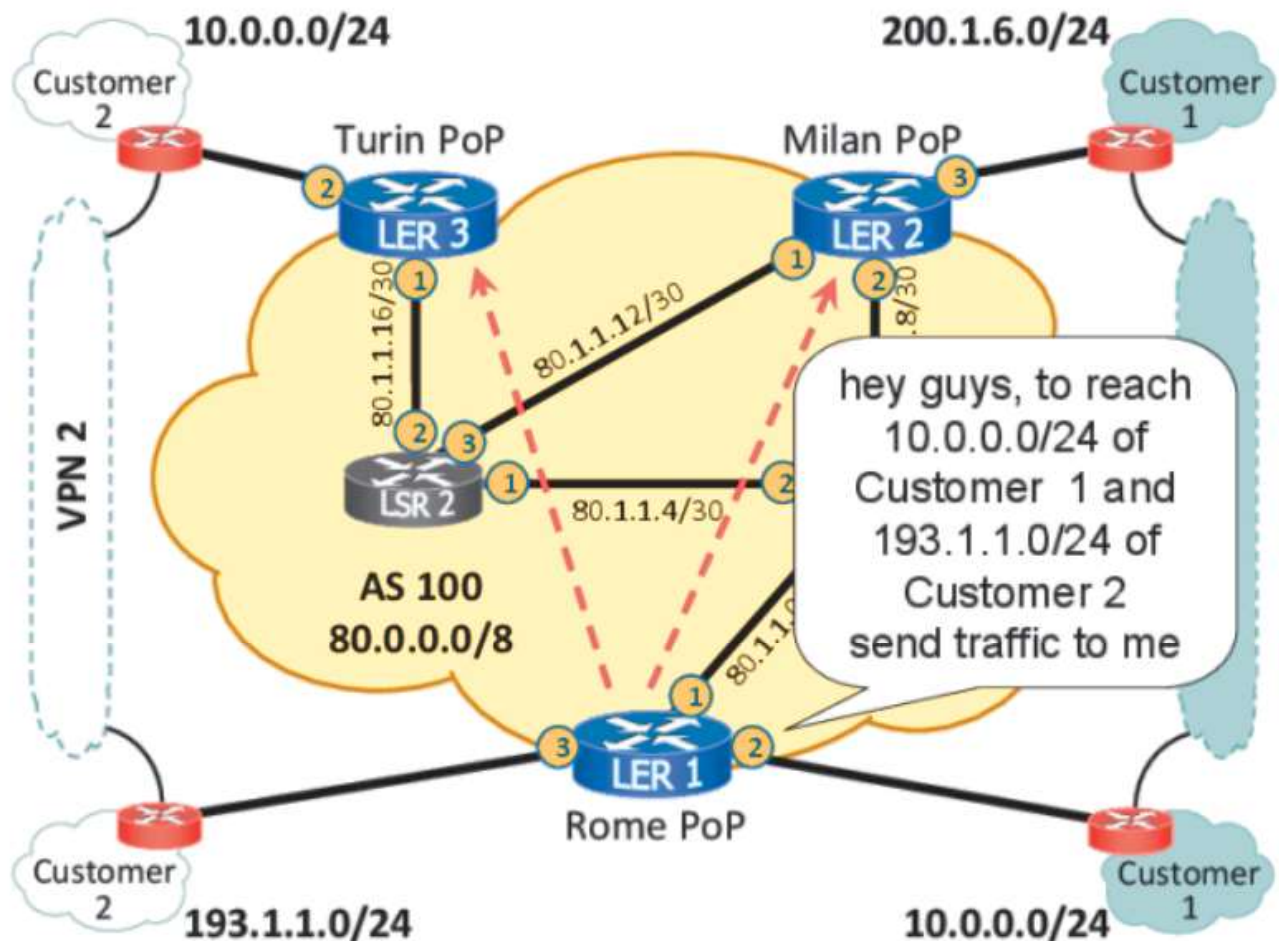
The CE routers belong to the customer, so they manage them, while the complexity of the VPN must reside in the PE routers. The CE doesn't have to configure or manage anything related to the VPN; they simply plug in the cable.

Move 1: Any-to-Any IP Connectivity Among PEs



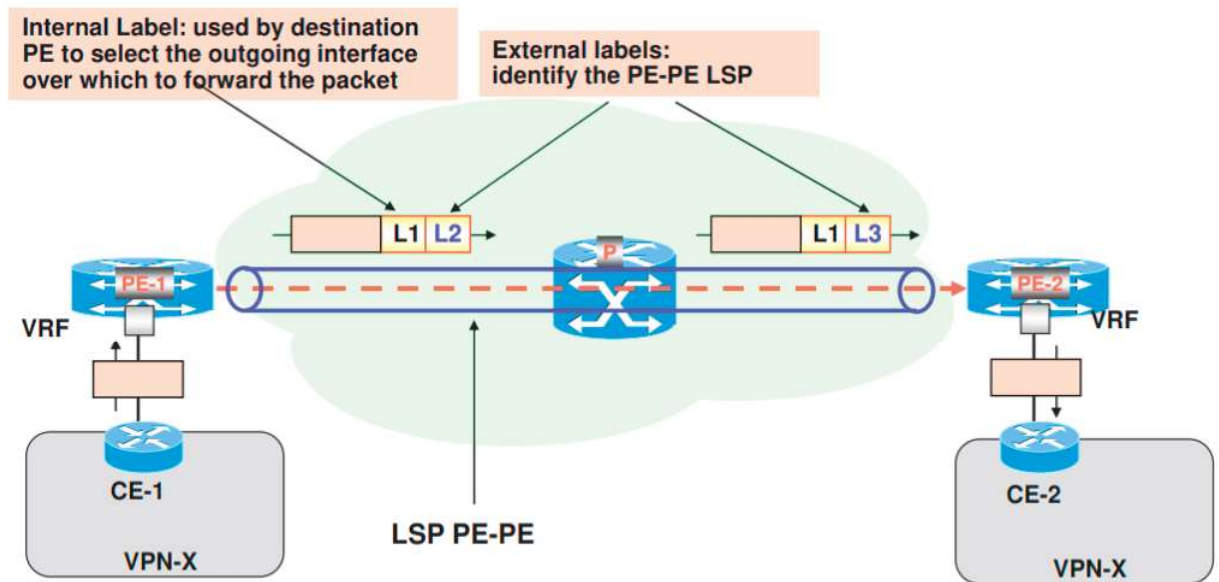
- Assign a loopback address to each PE router and use an IGP (e.g., OSPF or IS-IS) to announce these addresses as /32 prefixes, ensuring any-to-any connectivity among them.
- The loopback address serves as a static address that remains active even if failures occur (e.g., a network card dies).

Move 2: Use BGP to distribute customer prefixes

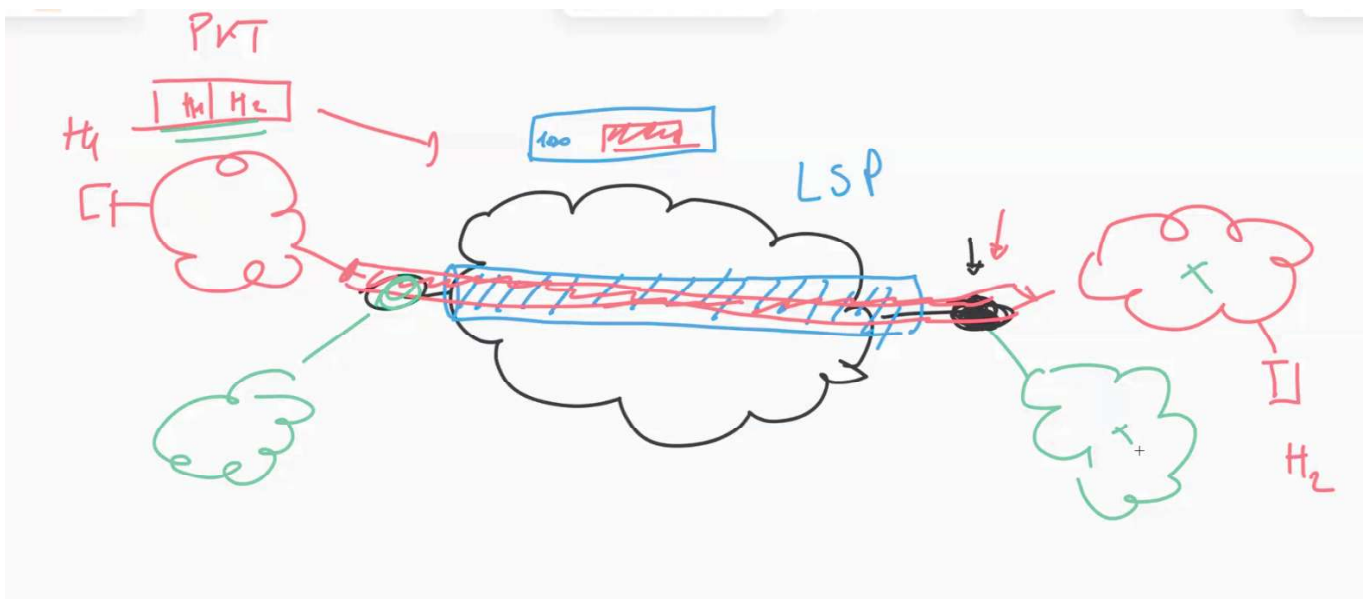


- **Multi-Protocol BGP (MP-BGP)** is used as a signaling protocol to distribute reachability information about customer prefixes.
- MP-BGP treats VPNs as a separate address family.
- The concept of the “customer” (i.e., the “L3VPN identifier”) is defined.
- PE routers establish a full mesh of iBGP peerings.
- A PE announces to all other PEs the customer prefixes it can reach via the CE router it is connected to.

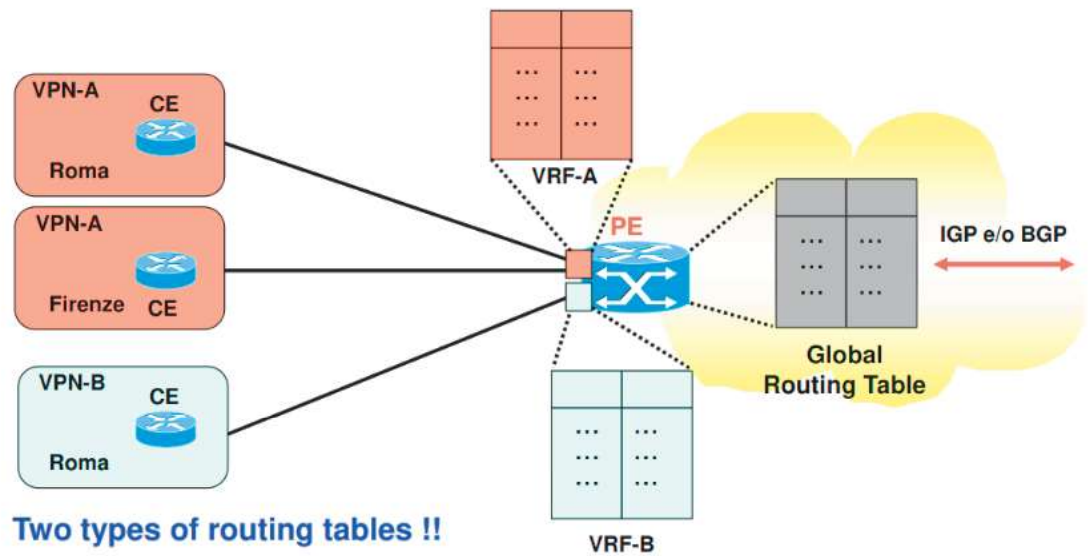
Move 3: Use MPLS encapsulation among PEs



- The PE-PE LSPs can be “Hop-by-Hop” or “Explicitly Routed”
- The PE-PE LSPs can be either **Hop-by-Hop** or **Explicitly Routed**.



Finally, to avoid conflicts when multiple VPNs operate on the same nodes, **VRF (Virtual Routing and Forwarding)** solves the problem. Each customer is given a separate routing table. These routing tables are called VRFs. When a packet comes from the interface connecting Customer 1 and the Rome node, the Rome node knows which VRF to inspect. VRFs are not necessarily one-to-one in terms of interfaces, but each customer has its own VRF.



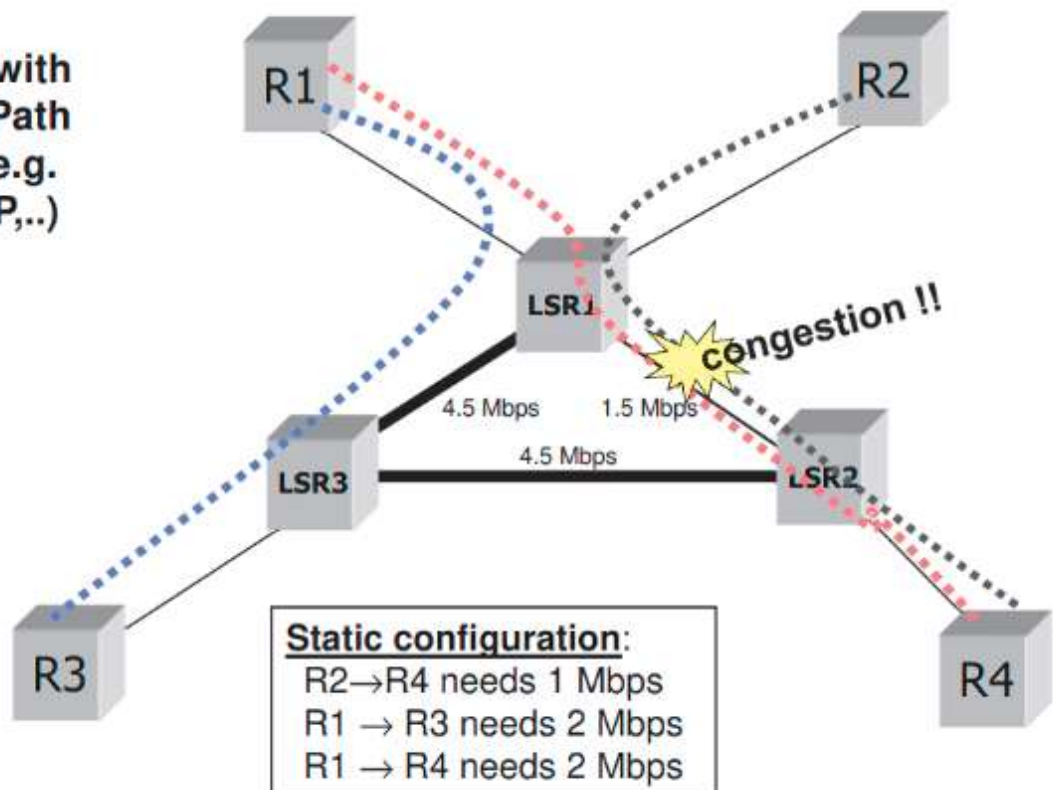
- Two types of routing tables:
 - Global routing table
 - VRF (VPN Routing & Forwarding)

LSP for Traffic Engineering

In the context of routing, finding a path from one point to another requires routing tables, where we install rules and run algorithms such as Dijkstra.

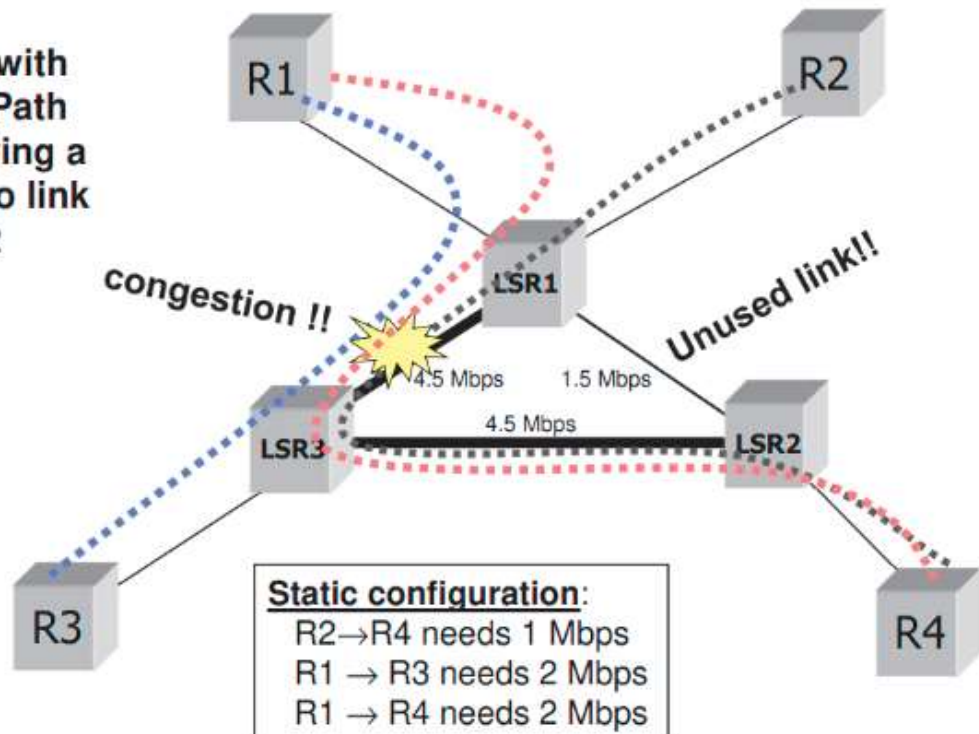
Dijkstra finds the shortest path, but sometimes the shortest path is not the best choice. For instance:

Scenario with Shortest Path routing (e.g. OSPF, RIP,..)



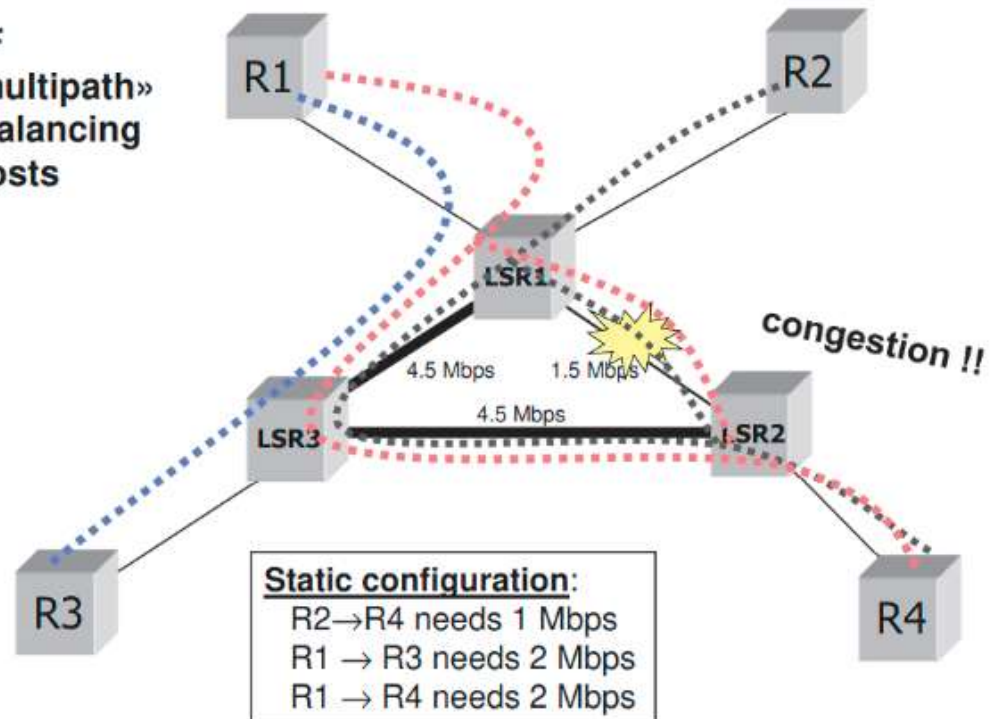
Dijkstra will solve congestion by simply increasing the weight of passing through a particular link, which shifts the congestion to another area.

Scenario with Shortest Path routing giving a high cost to link R1-R2



Splitting traffic into two paths also creates congestion.

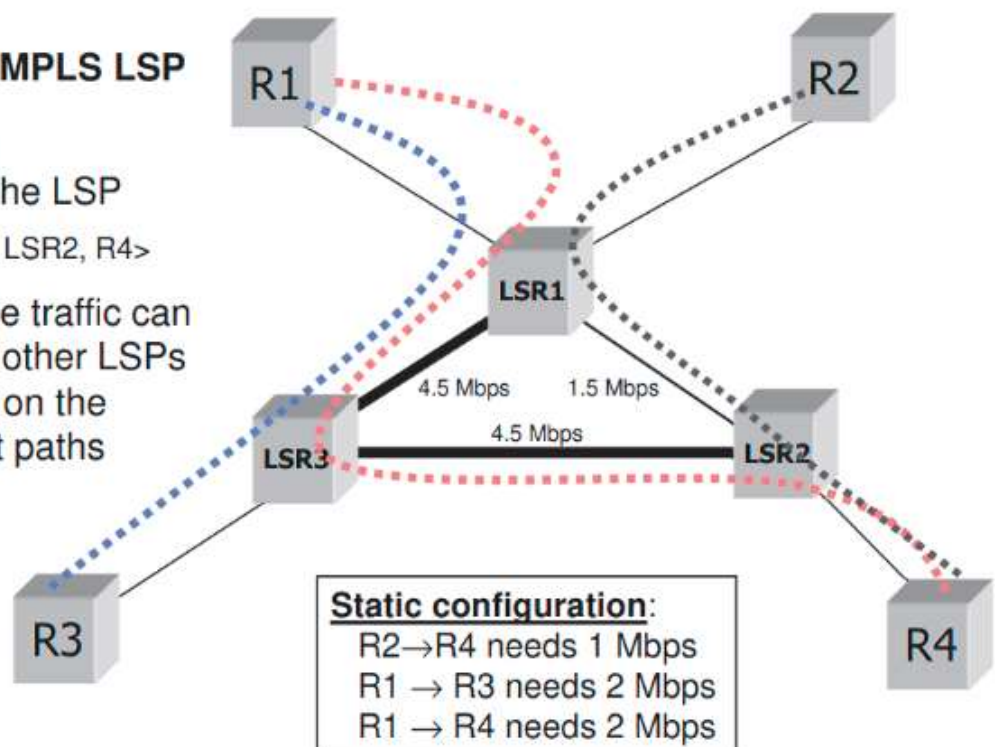
OSPF
«Equal Cost multipath»
with proper balancing
of link costs



We need a technology that supports **Traffic Engineering**—a function that allows network operators to balance the load. MPLS can state, for example, that only the red flow is allowed to pass through a specific link, effectively creating an LSP (Label Switched Path).

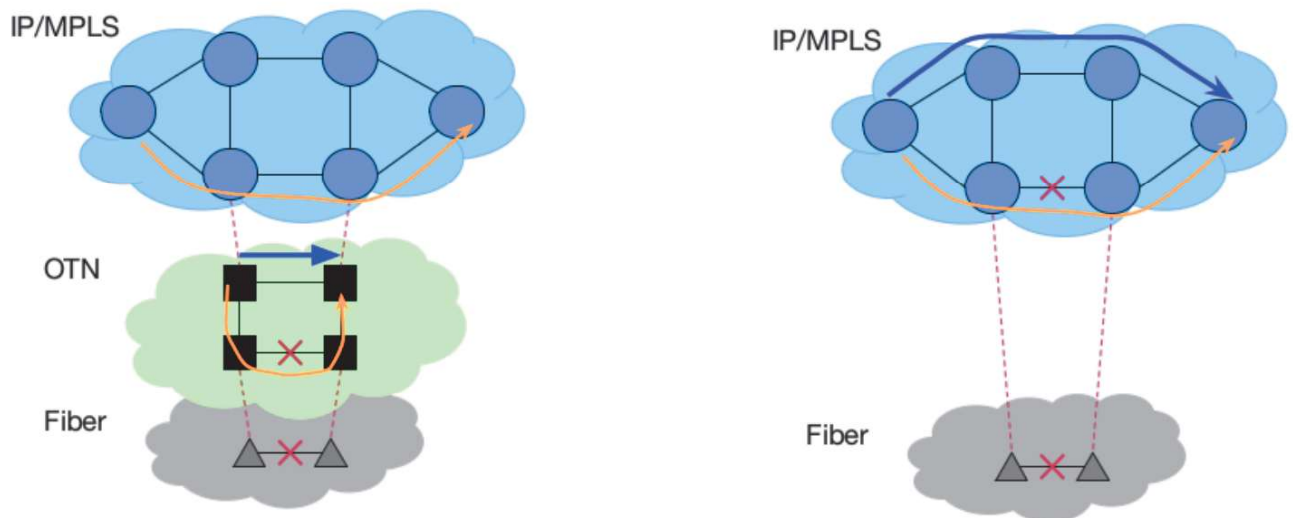
Scenario with MPLS LSP

- R1→R4 traffic is tunneled in the LSP
<R1, LSR1, LSR3, LSR2, R4>
- the resto of the traffic can be assigned to other LSPs or it can be left on the default shortest paths

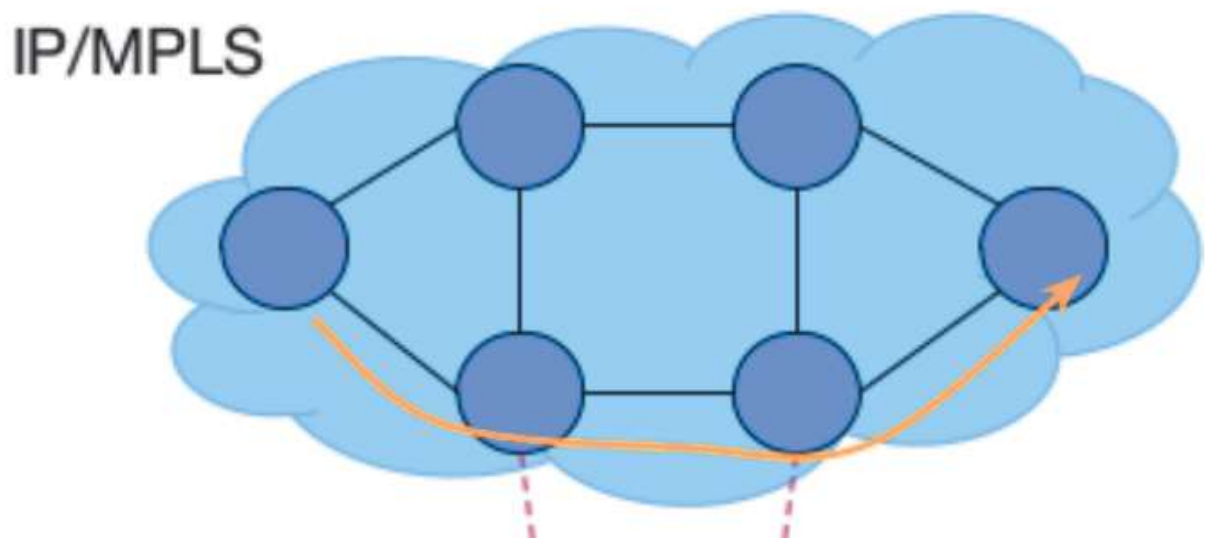


Fast Recovery after Failure Service

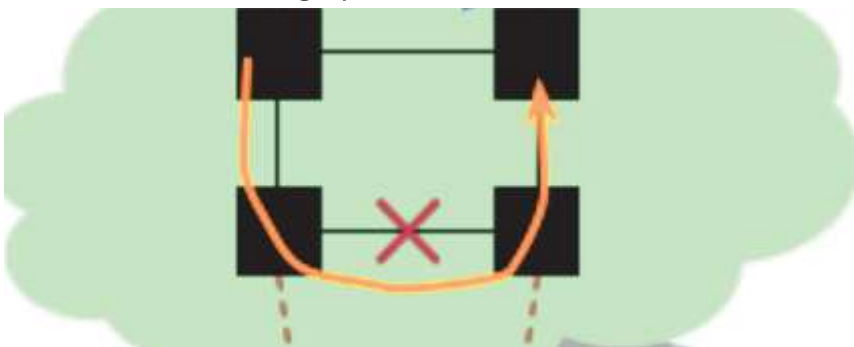
Fast Recovery after Failure Service: how does the IP net react to the failures? The net will find a new path but it takes time. So the provider can provide fast restoration, in the order of milliseconds, thanks to MPLS. Let's consider this example:



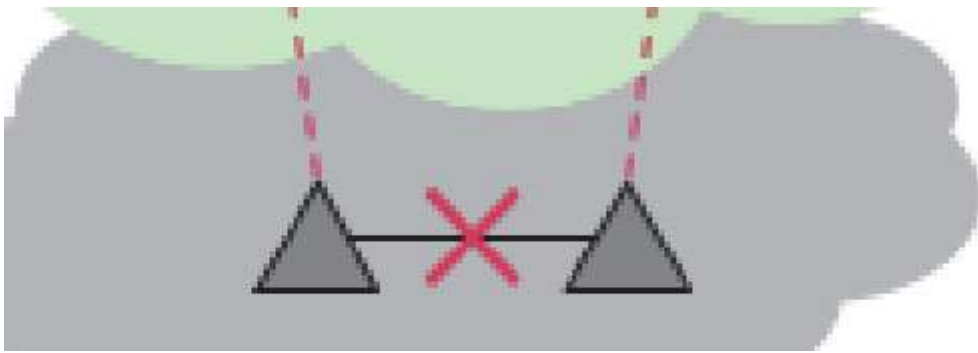
This IP/MPLS network:



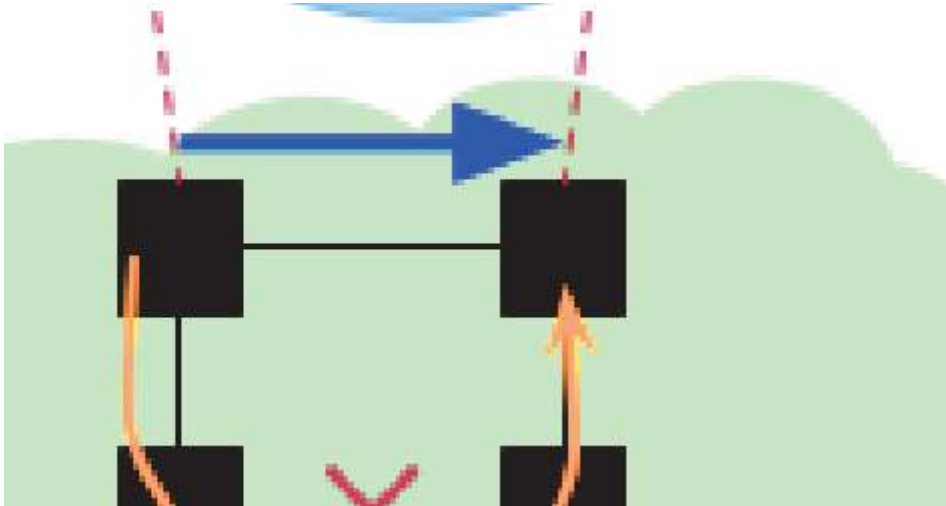
is created over this lightpath:



Let's assume this fiber fails:

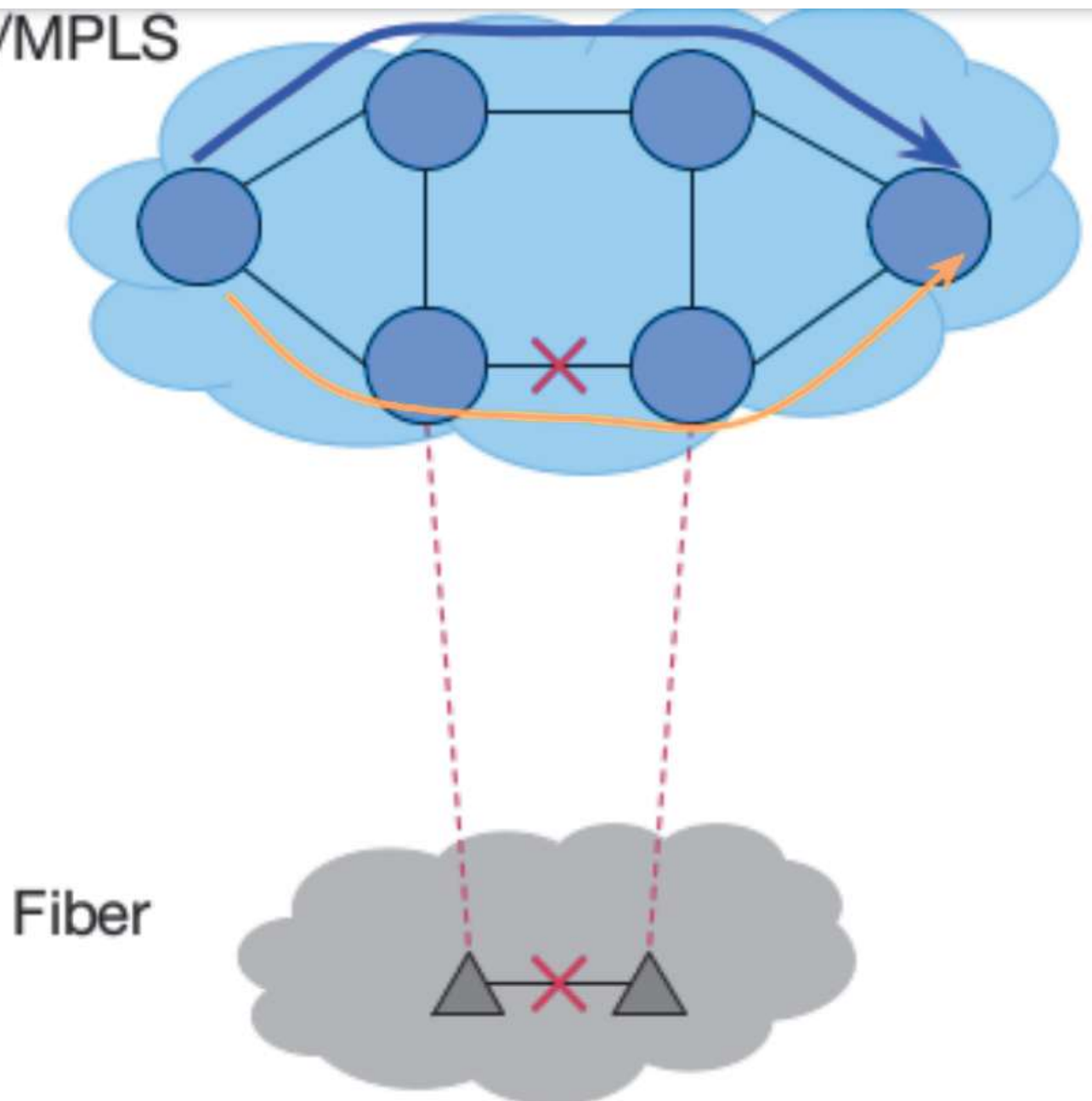


After detecting the failure, I find that I can pass here (blue arrow)



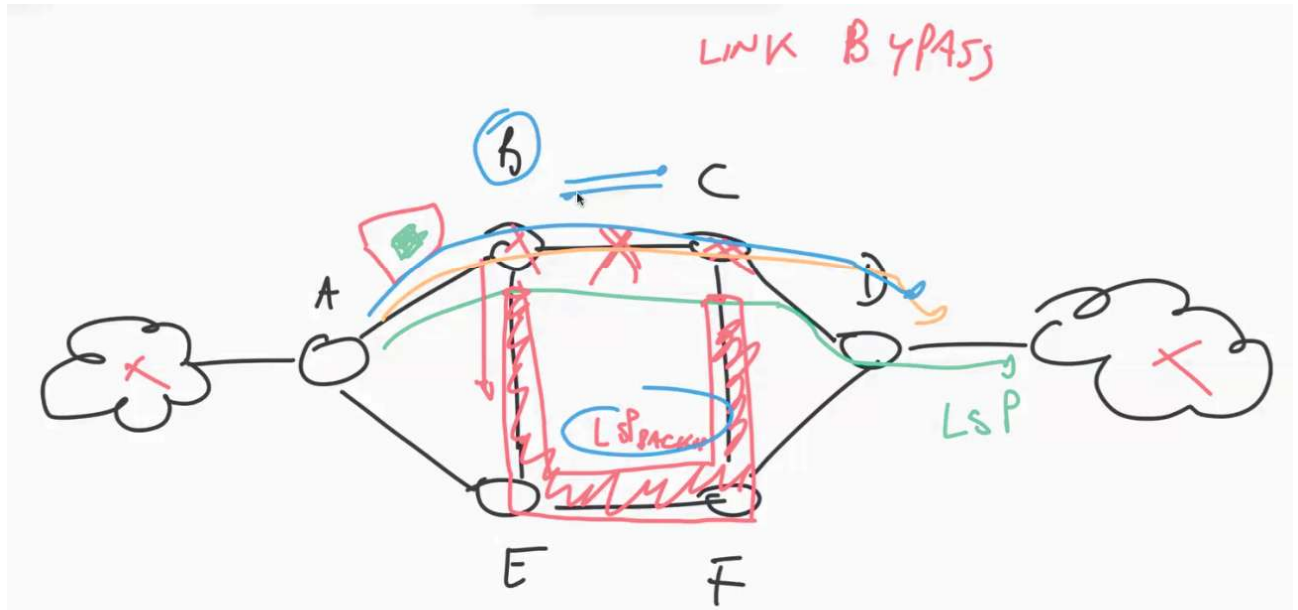
It would be nice to solve this problem on a higher level, the IP level.

IP/MPLS

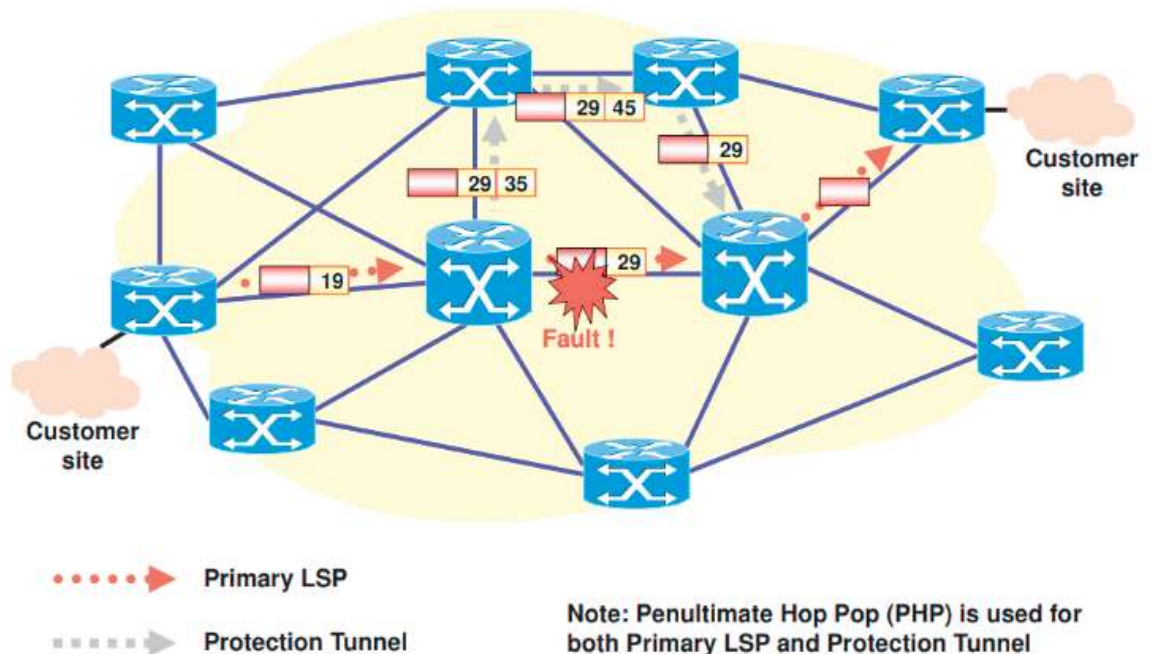


This way I don't have to wait until the optical network has converged, because I can find the alternative path directly in the IP net. How to implement this? We have two options:

- Fast Rerouting:



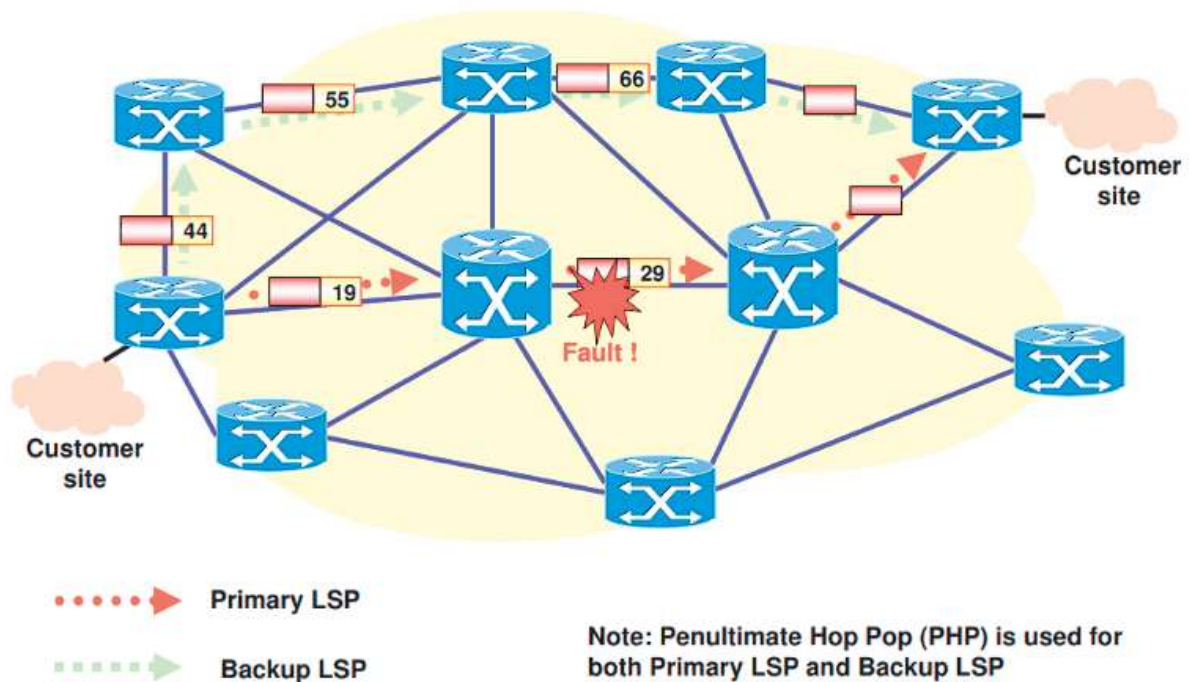
Fast rerouting – link protection



- Create a backup path to protect the data from the failed link. When a nearby node detects the failure, it encapsulates incoming packets so they can bypass the failed link and travel on the backup path.
- If there were different LSPs (Label Switched Paths) originally traveling through the failed link, all of them would be redirected to the backup path.
- A node can detect a failure through a "Hello-based" protocol, such as **Bidirectional Forwarding Detection (BFD)**, which identifies a failure when the continuously sent Hello messages stop arriving at the neighboring node.
- There are two reasons why this process is faster than the normal Plain Control:

1. We don't need to notify all nodes that the link has failed; only the node before the failure, which ultimately performs the encapsulation, needs to know.
 2. No computation is required, as everything is pre-configured ahead of time.
- The downside of this method is that the total number of hops in the path is increased, while the next technique avoids this problem.
 - **Backup LSP:**

backup LSP – path protection



- Instead of using *link-protection*, we implement *path-protection*.
- When a failure is detected, the data flow is rerouted over a completely different end-to-end backup LSP.
- This approach increases performance (same number of hops) but comes at a higher cost, as creating a backup LSP for each primary LSP is more expensive than protecting only the link.

Wooclap Solutions

1 - Put in the correct order the different phases of the circuit setup procedure:

1. The source node ask to the network the creation of a new circuit.
2. The Network compute a path from the source to the destination node
3. the network checks if, in each link of the path, there is an available sub-channel
4. new rules are inserted in the switching table of the network nodes

5. the clients are notified that the circuit is up
6. the communication happens
7. the network monitors the circuit
8. the network tears down the circuit

2- Match each of the listed features with the correct optical network generation

First Generation Optical Network:

- Switching is performed in the electrical domain
- It Exploits only the EM properties of the optical fibers to improve the performance.

Second Generation Optical Network:

- The optical domain is used to create point o point link links between clients devices
- supports routing and switching in the optical domain
- it uses meshed topology
 - (Note: *It's because in the first generation we use optics only to create point to point links, so the topology of the first generation is "point-to-point"*)
- Fully exploit the high bandwidth of the optical fibers.

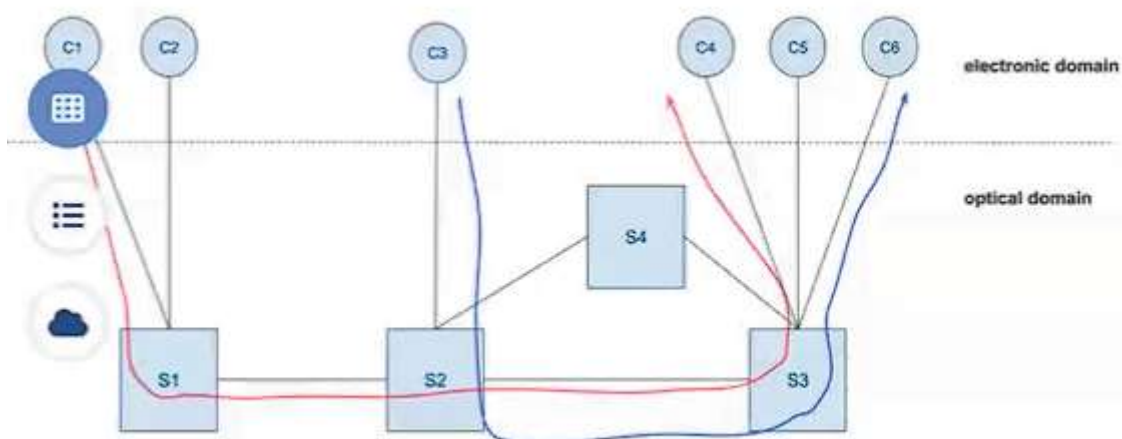
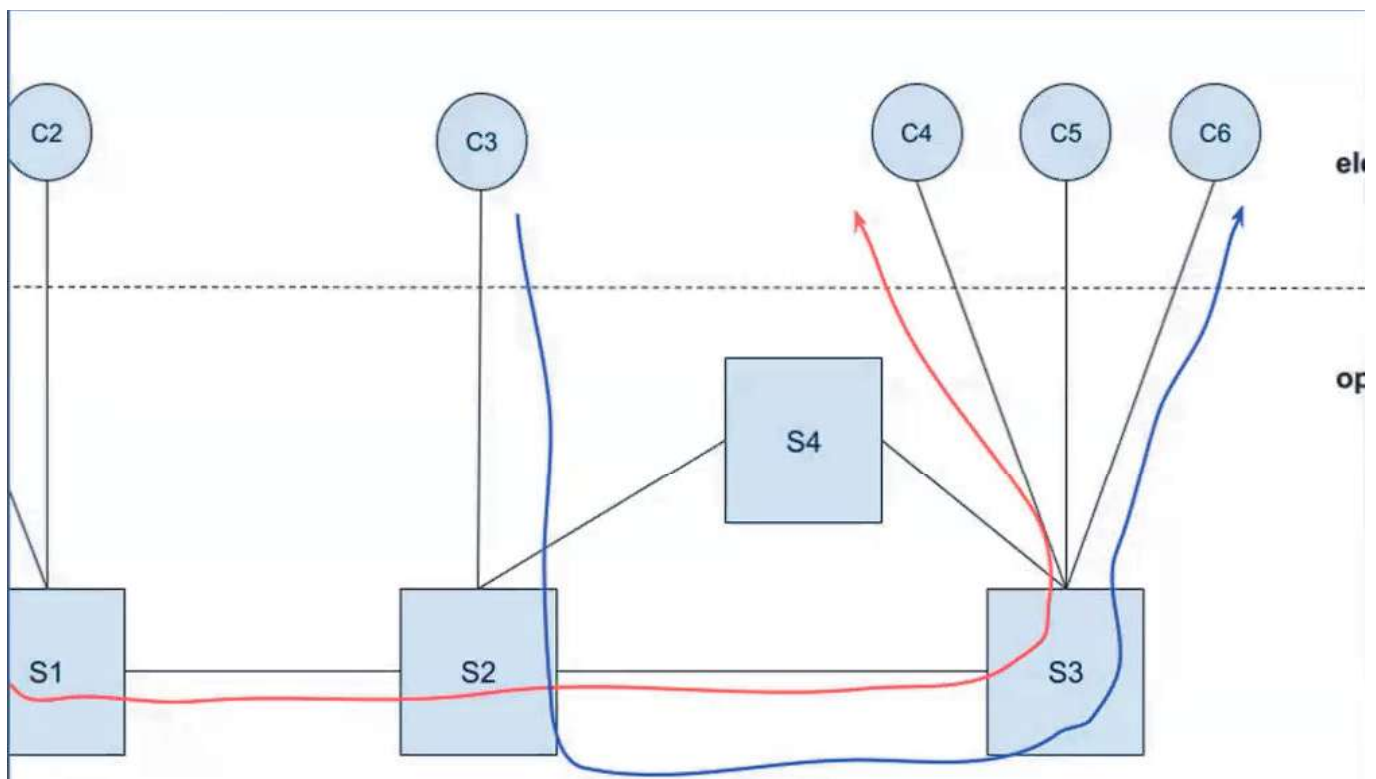
3- Match the devices and functionalities

- make cost effective the handling of passthrough traffic: OADM
- allows the switching of optical signal at a large scale: OXC
- is the termination of an optical link: OLT

4- Consider the All Optical Network shown in the figure, where each client is equipped with a transponder able to transmit at a single lambda (red or blue) and whose switching elements are able to handle only these two wavelengths.

Answer the following questions:

1. can a light-path be established between clients C2 and C5?
2. eventually, what color can be used for this new connection?
3. now assume that two new clients (C7 and C8) are connected at the ON, through the switches S1 and S4 respectively. Can the network support the creation of a light-path between the two clients.



5-Find the correct match for the layers of the OTN stack

- it represents the logical connection between the client's interfaces: **ODU**
- at this layer we have visibility of Optical Amplifiers: **OTS**
- It is the portion of the network between two multiplexers: **OMS**
- It is the only layer in the optical domain where we have access at the single wavelengths: **OCh**
- it represents a logical connection between two end points in an optical sub-network: **OTU**

6- Assuming that a failure happens in the optical sub-network 1, which type of Defect Indicator allows the client C2 to stop raising its alarm?

Figure missing, it's not that relevant since the only layer the client see is ODU

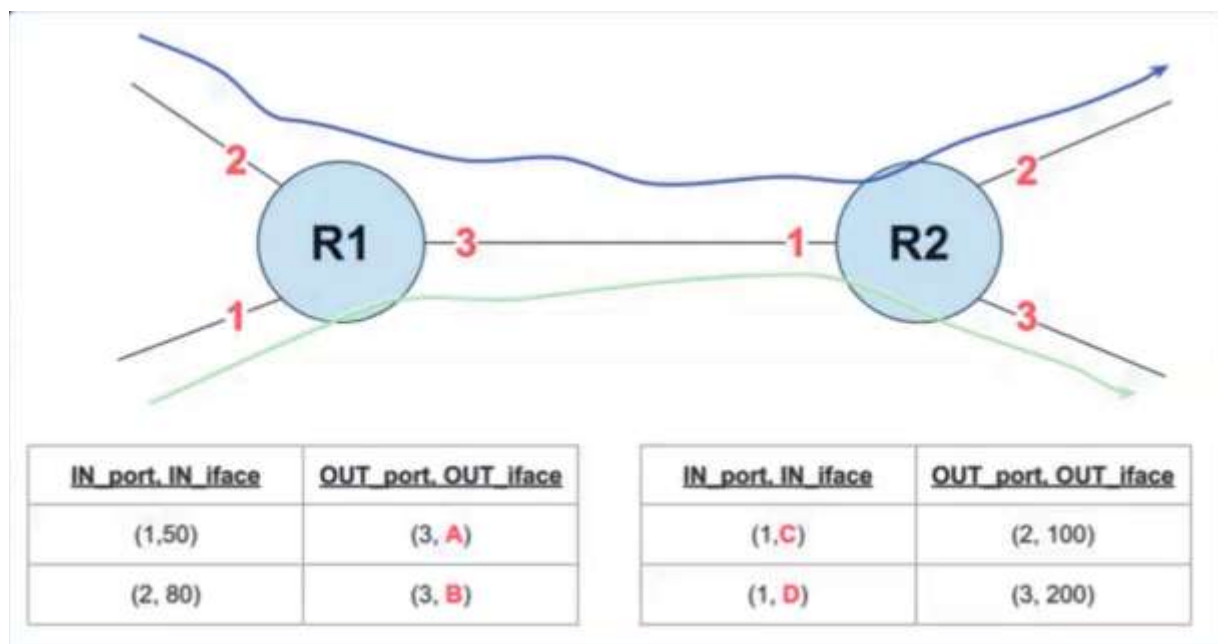
- ✓ ODU-FDI
- ✗ OTU-FDI
- ✗ OCh-FDI
- ✗ OMS-FDI

7- Indicate what type of overhead is used to carry the trace information

- fiber link: **Optical Supervisory Channel**
- optical channel: **Pilote Tone**
- light-path: **Rate Preserving Channel**

8 - Considering the situation shown in the figure, which one of the following statement is true?

1. ✗ we can assign A, B, C and D whatever value
2. ✗ we can assign to A and B whatever value, then D=A and C=B
3. ✗ we can assign to A and B whatever value, then D=B and C=A
4. ✓ none of the above



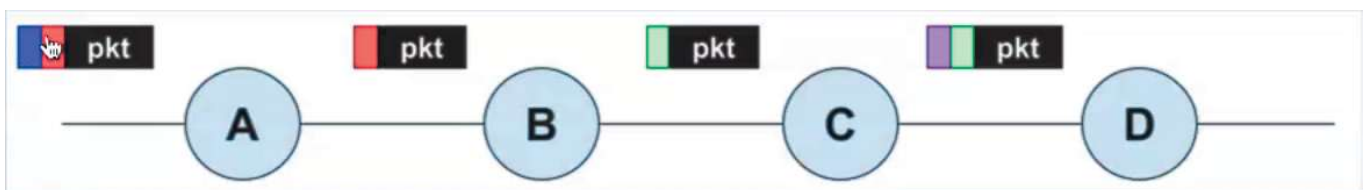
This is because MPLS does not admit two packets with the same label on the same link.
"Whatever Value means also the possibility of the same value for both the packets."

9 - Fill the white spaces with MPLS or IP

- The ① dataplane allows the packet forwarding only over the shortest paths;
- ② allows for a more flexible routing and switching than ③ ;
- ④ is a connectionless environment, while ⑤ requires the use of a tunnel setup procedure;

1. IP
2. MPLS
3. IP
4. IP
5. MPLS

10 - Considering the example shown in the figure, match each node with the MPLS action it is performing:



- A: PUSH
- B: SWAP
- C: POP